

SUPPLEMENT

Interleukin-6 is an effective mediator of therapeutic efficacy in acute respiratory distress syndrome: an individual patient data meta-analysis of RCTs

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Supplement A

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Table S1: Brief descriptions of the studies.

Table S2: Baseline characteristics of trial participants included in the present analyses.

Table S3: Joint modeling results for each trial, adjusted for age, sex, BMI, and SOFA or APACHE II scores.

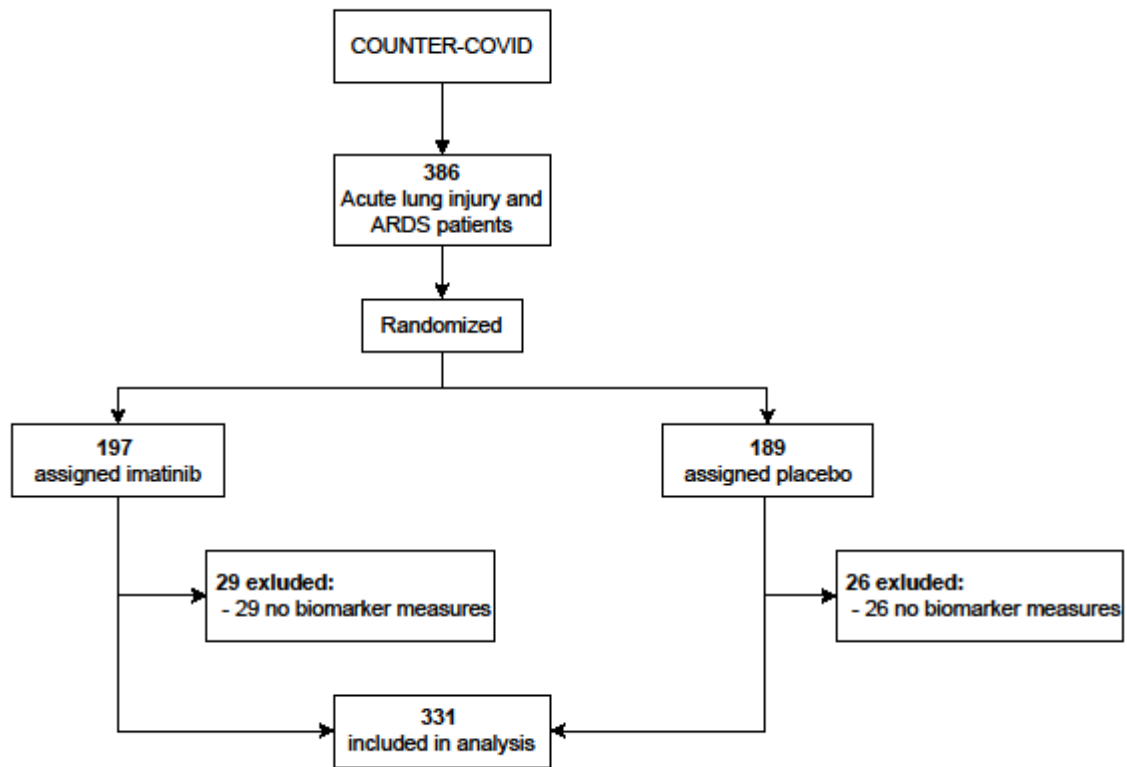
Table S4: Comparison of patient baseline characteristics with no IL-6 observations to patients included in the present study.

Table S5: Missing IL-6 observations and deaths in IL-6 observation periods.

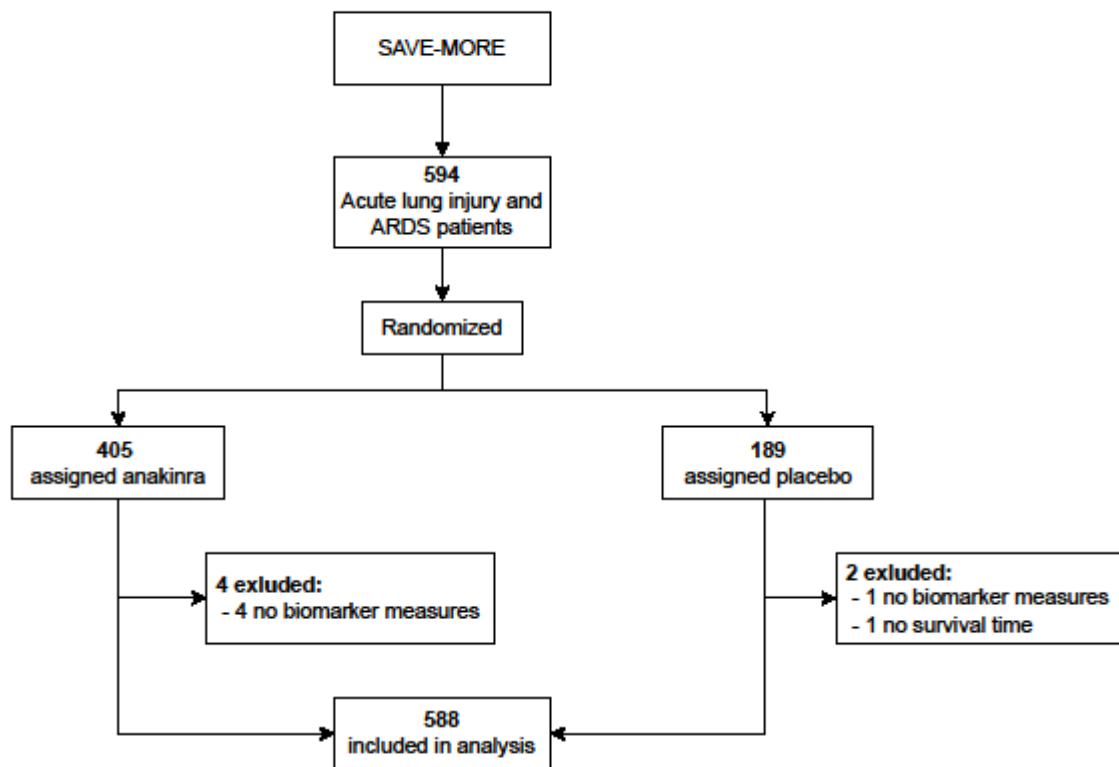
Supplement B includes a detailed description of the statistical analysis approach.

Figure S1: Flowcharts of patient inclusions per trial.

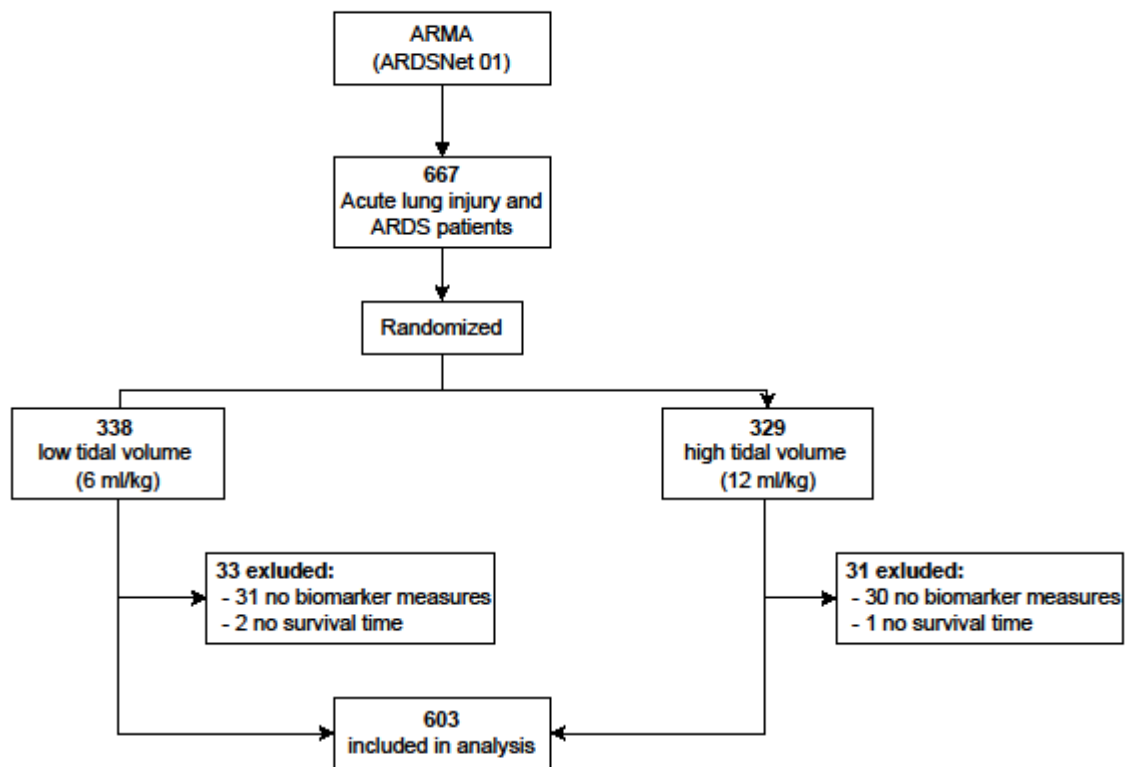
A: COUNTER-COVID



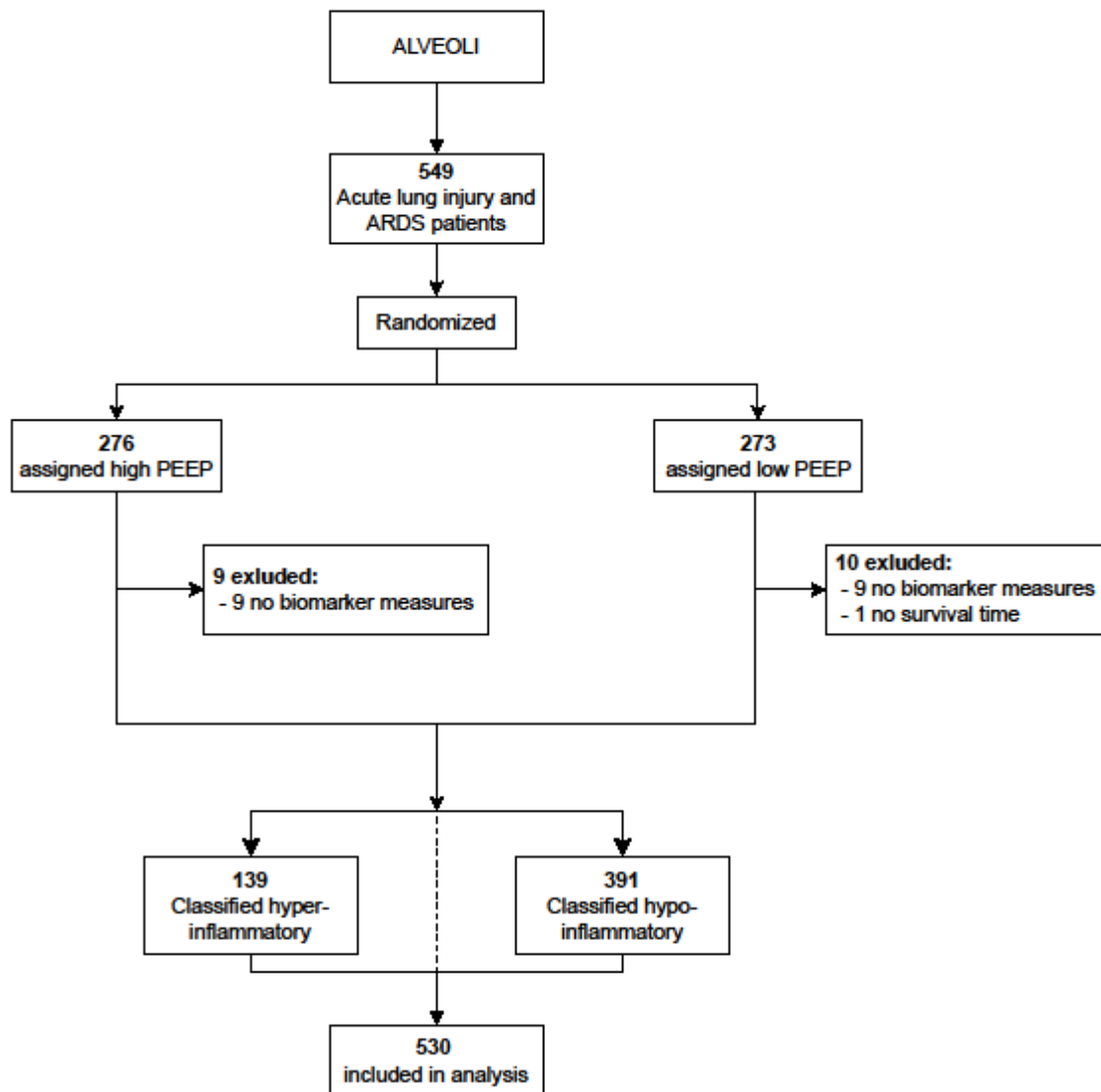
B: SAVE-MORE



C: ARMA



D: ALVEOLI



E: HARP-2

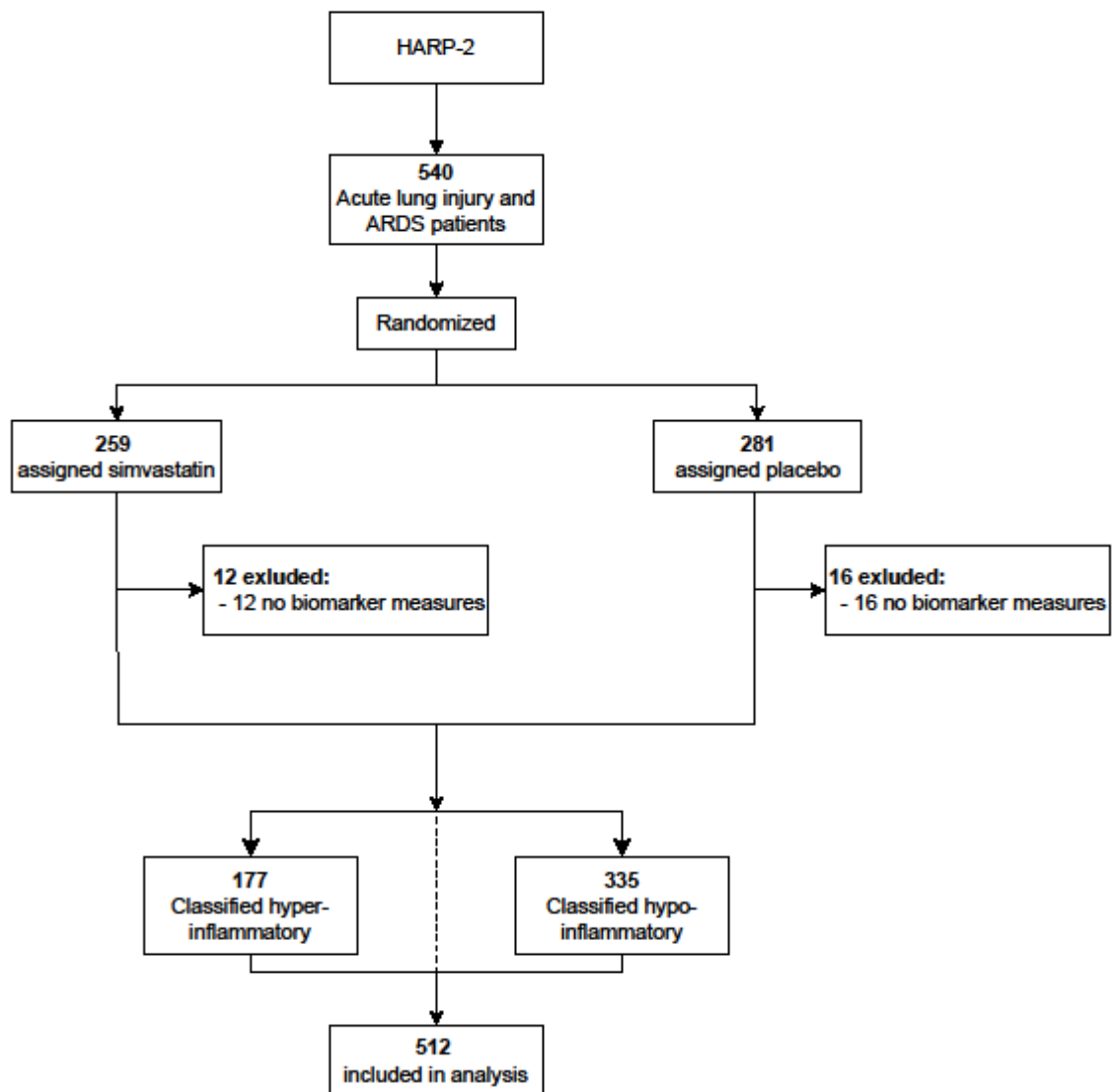
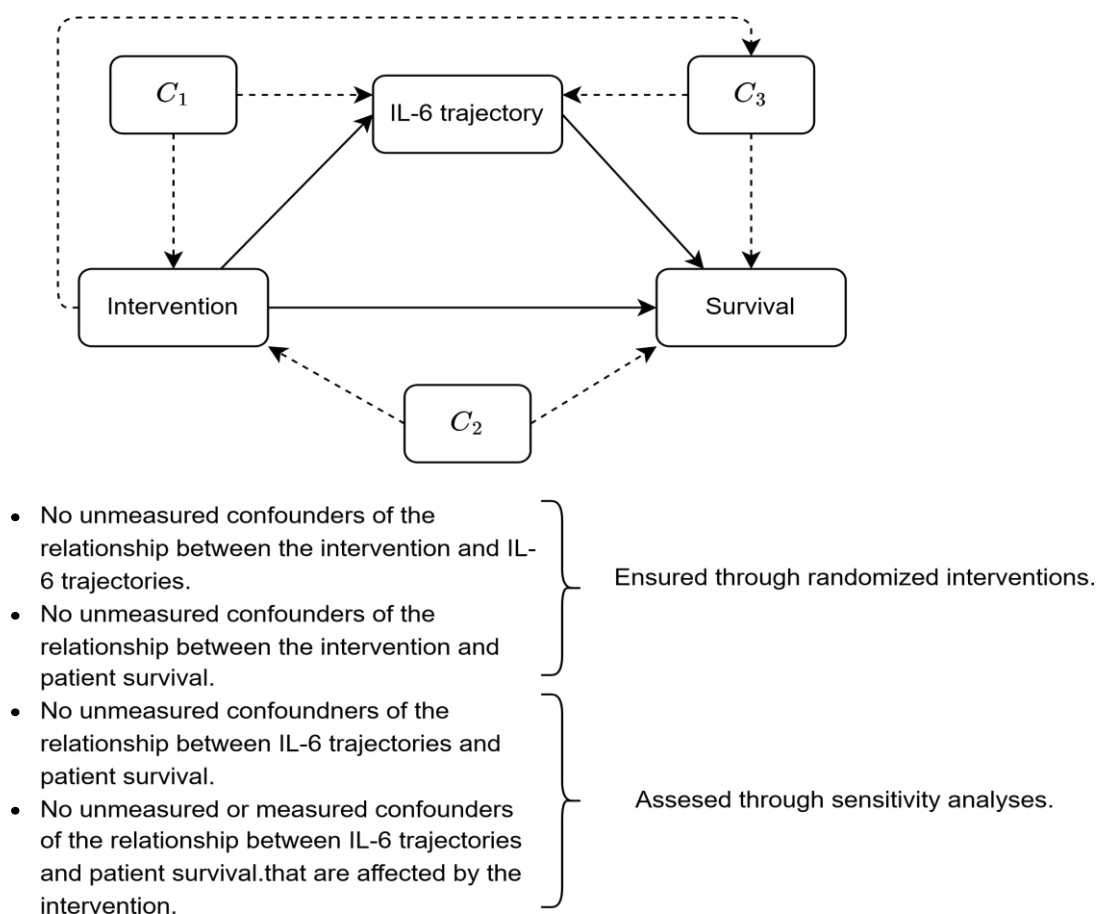
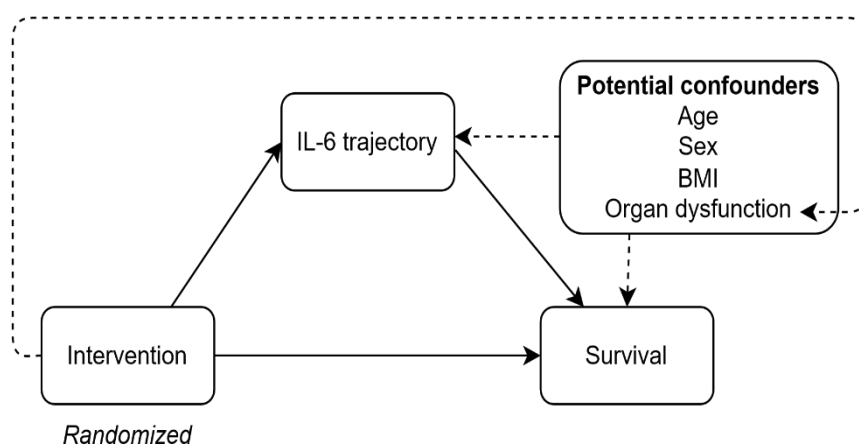


Figure S2: Graphical representation of confounding assumptions.

A: Confounding assumptions for mediation analysis.



B: Possible mediator-outcome confounding in present analysis.



In our analysis, we include covariates age, sex, BMI, and measures of organ dysfunction (SOFA or APACHE-2, depending on availability in each trial data set) at baseline. In COUNTER-COVID, we adjust for respiratory rate because SOFA and APACHE scores were unavailable. As illustrated in B, organ dysfunction after randomization may be affected by interventions.

Figure S3: Total, direct, and indirect effects per intervention (and per inflammatory sub-phenotype) for 90-day endpoint. Adjusted for age, sex, BMI, and either respiratory rate, SOFA or APACHE II/III scores or respiratory rate at baseline per intervention (and per inflammatory sub-phenotype) for 90-day survival.

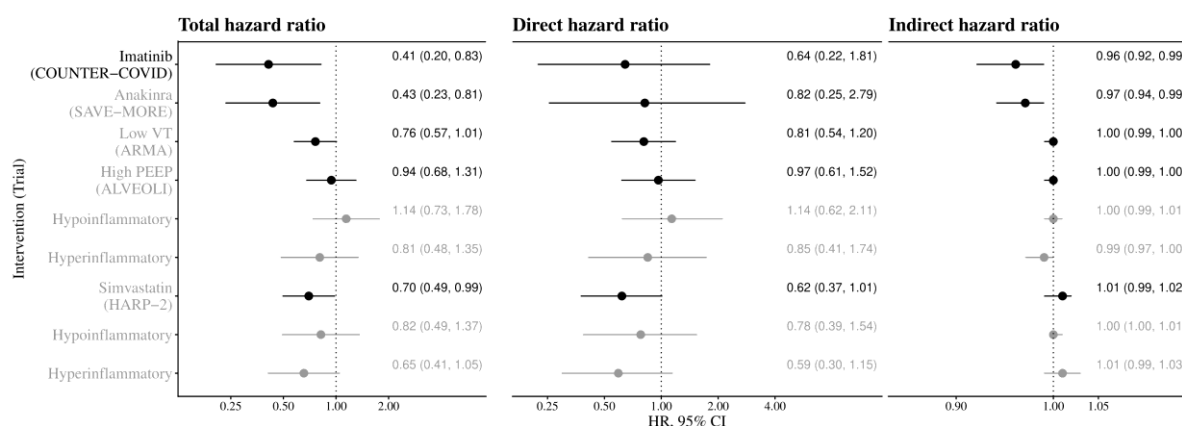


Figure S4: Meta-analysis results for the association between IL-6 concentration and mortality for 90-day endpoint.

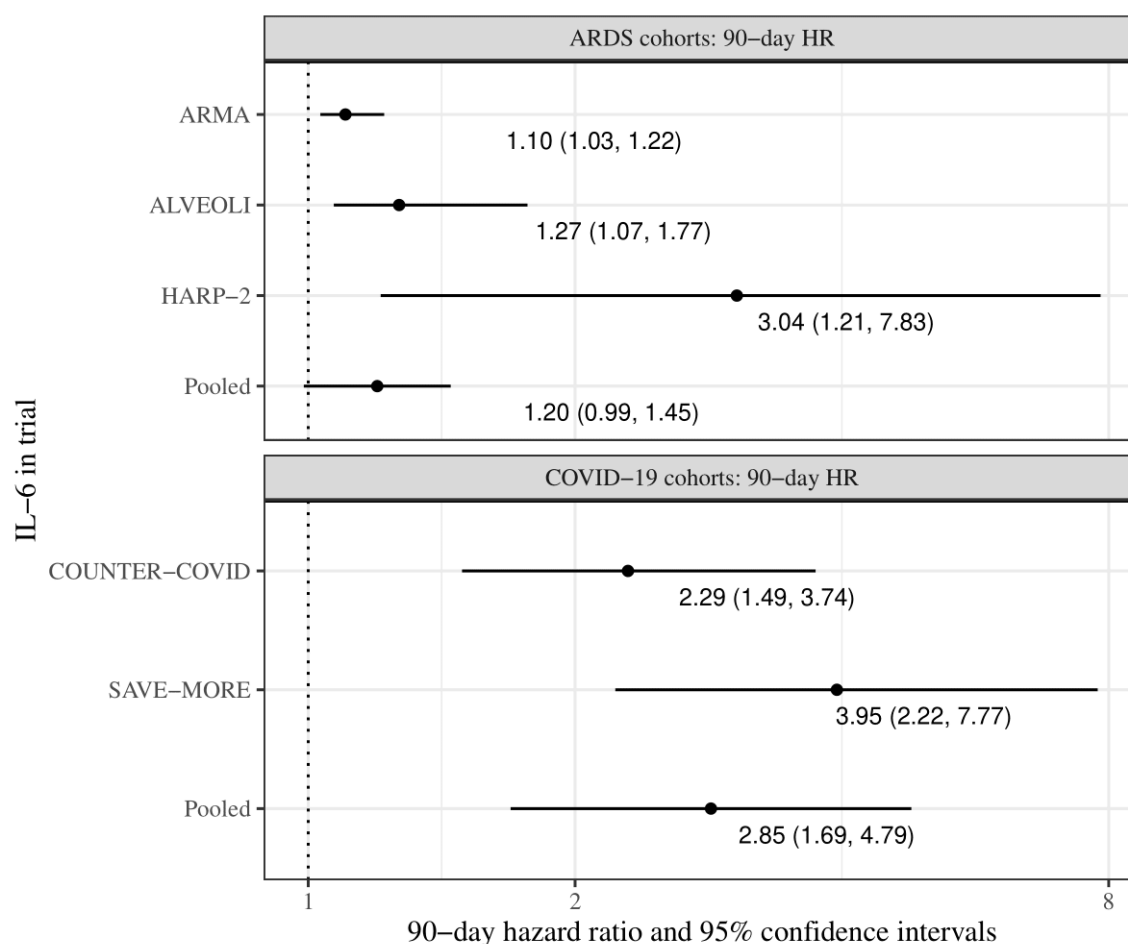
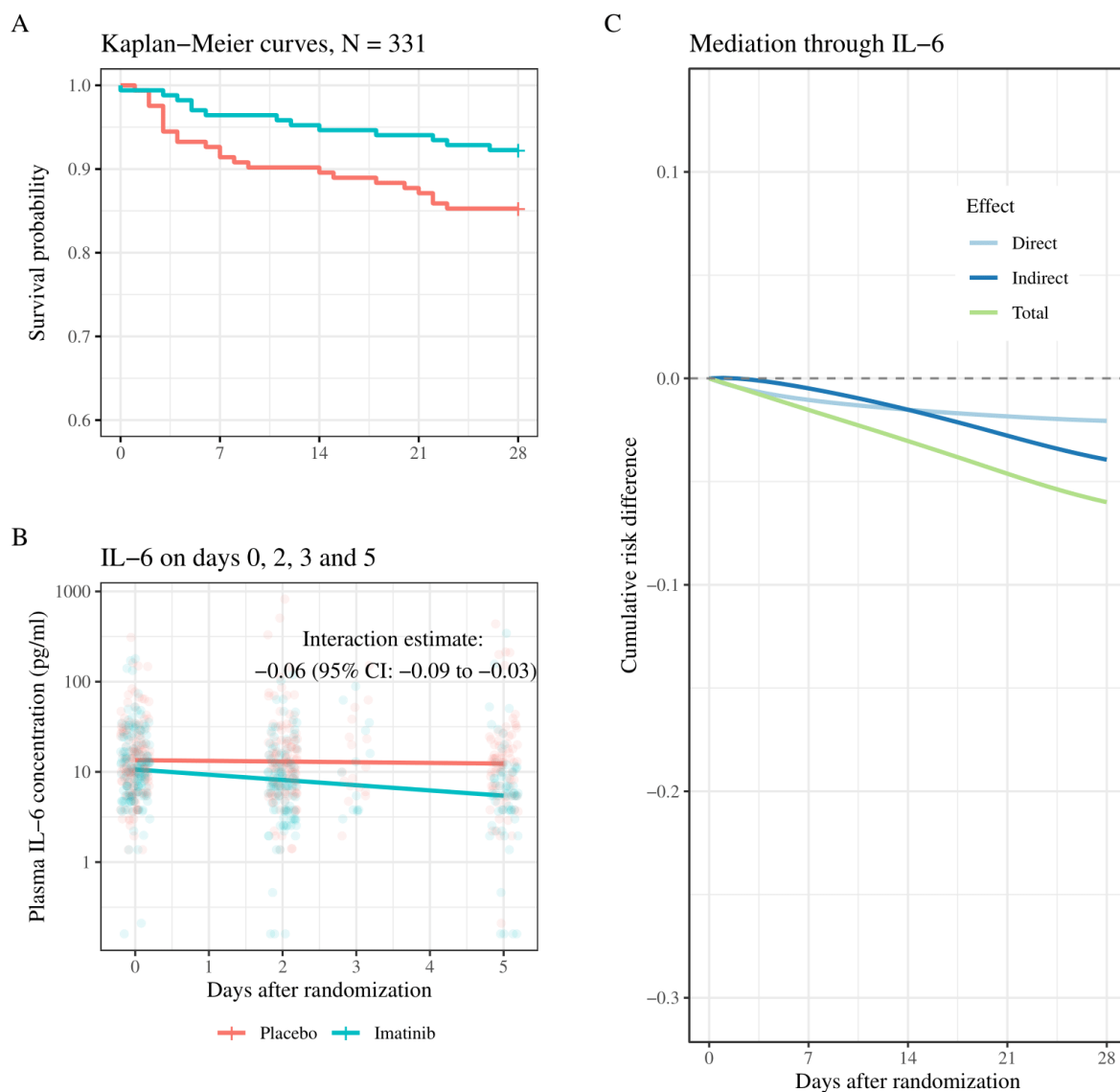


Figure S5: Kaplan-Meier curves (panel A), IL-6 trajectories (panel B) and time-dependent direct, indirect, and total effects with IL-6 as a potential mediator (panel C) for each trial. The interaction estimates in panels B are the interactions between intervention and time, obtained from the joint models. The time-dependent direct, indirect, and total effects are presented on a risk difference scale. For each patient, we used the joint models to estimate the cumulative incidence functions under intervention and control. The cumulative risk differences between intervention and control are depicted as total effects. For each patient, we then estimated the effect of intervening only on the survival outcome (hypothetically leaving IL-6 unaffected by treatment). The risk differences between these estimates of intervening on the survival outcome versus not intervening on the survival outcome are the direct effects of the interventions. To obtain estimates of the indirect effects on a risk difference scale, we estimated the effects of hypothetically intervening on only the mediator.

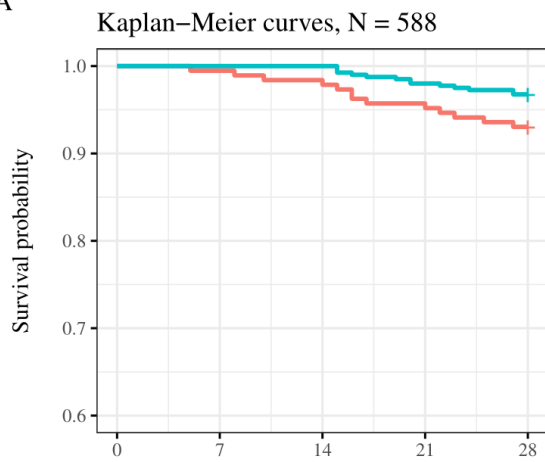
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COUNTER-COVID

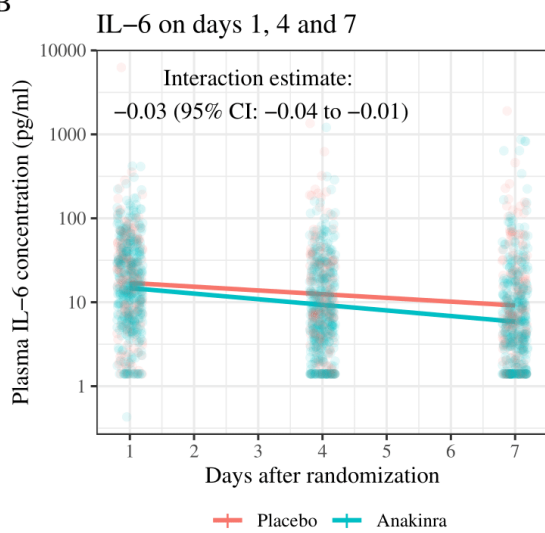


II: SAVE-MORE
SAVE-MORE

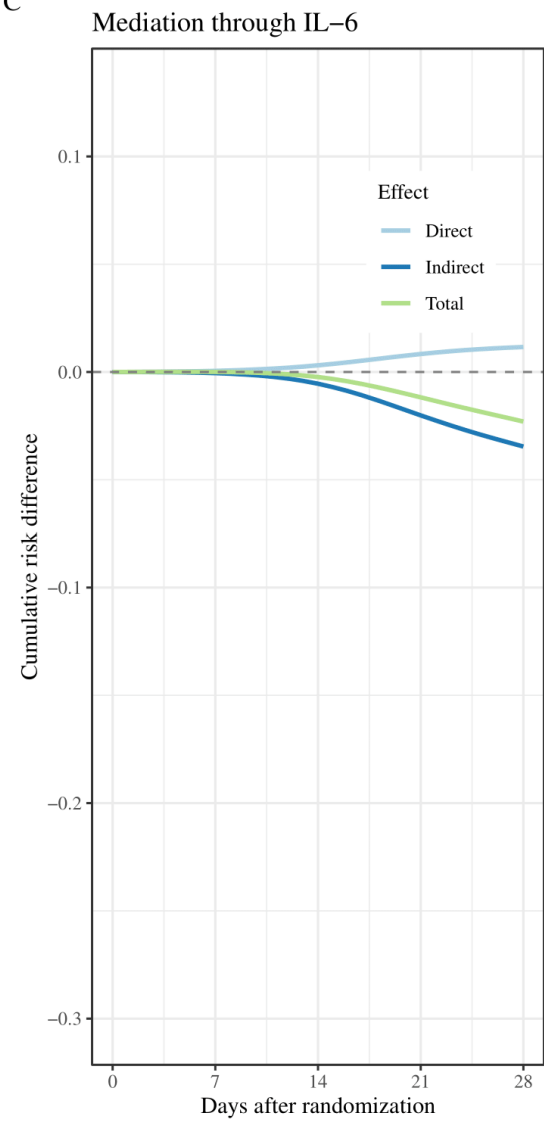
A



B



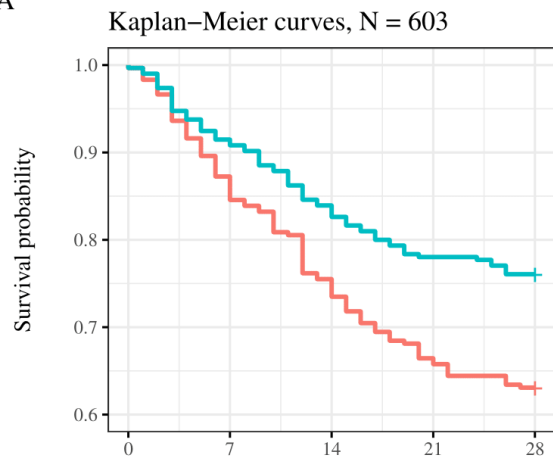
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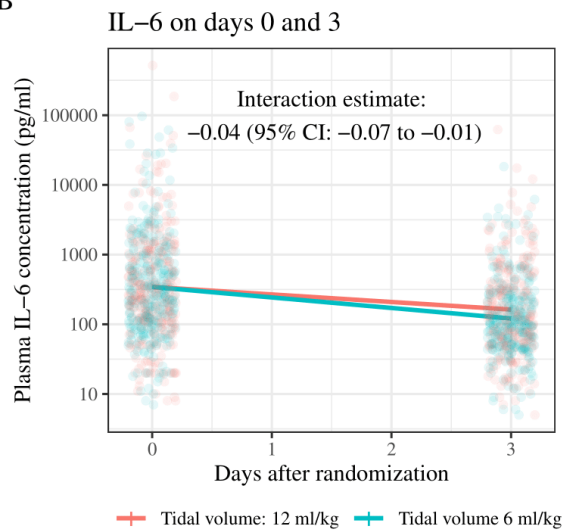
III: ARMA

ARMA

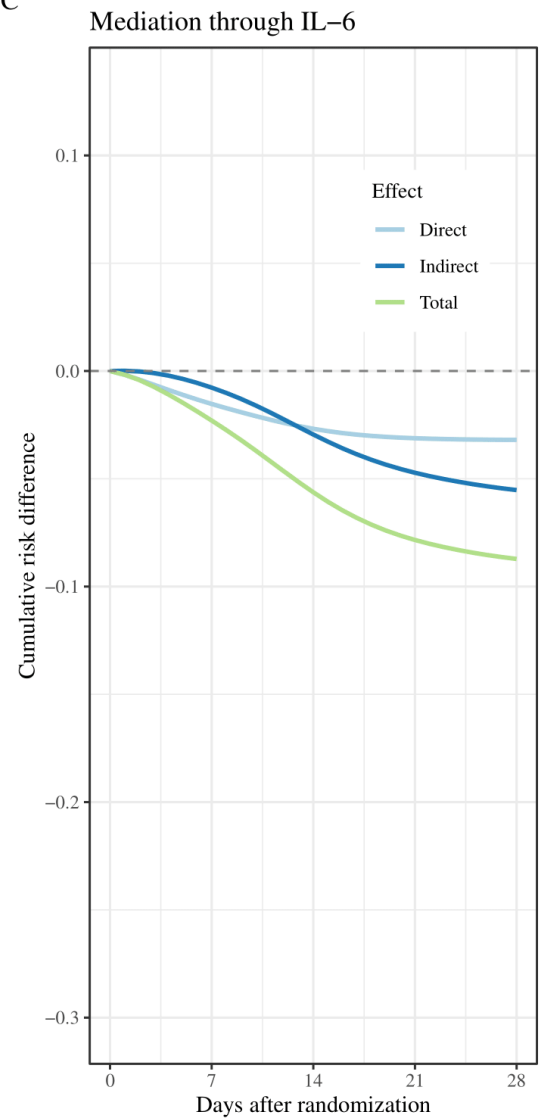
A



B



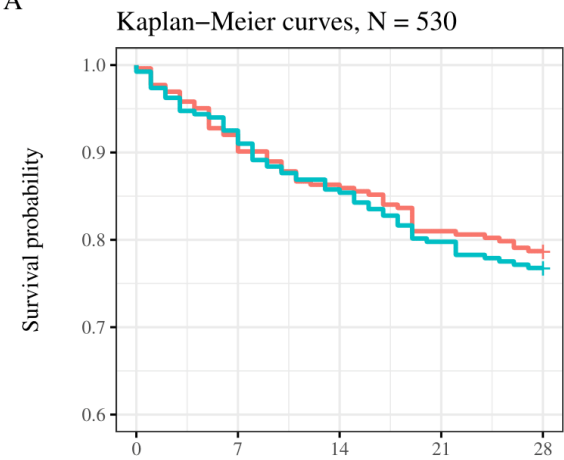
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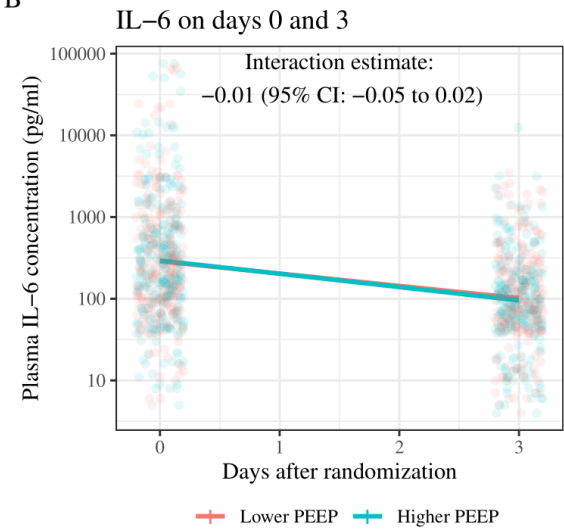
IV: ALVEOLI (all patients)

ALVEOLI

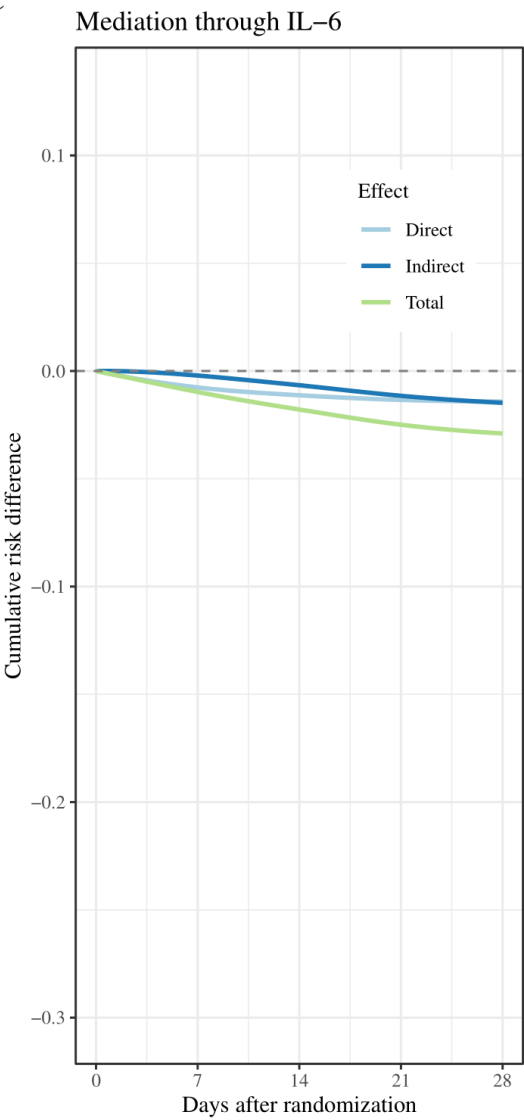
A



B



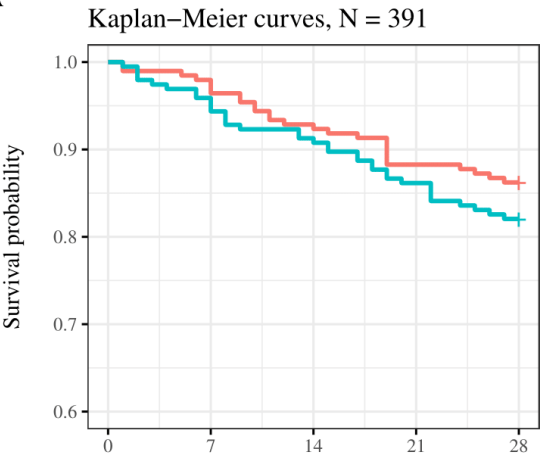
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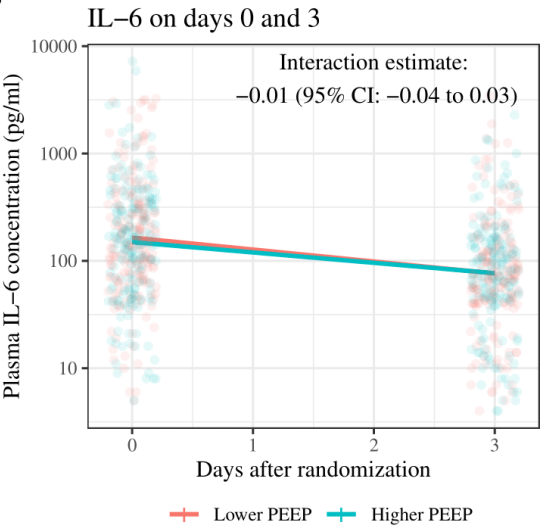
V: ALVEOLI (hypoinflammatory)

ALVEOLI (hypoinflammatory)

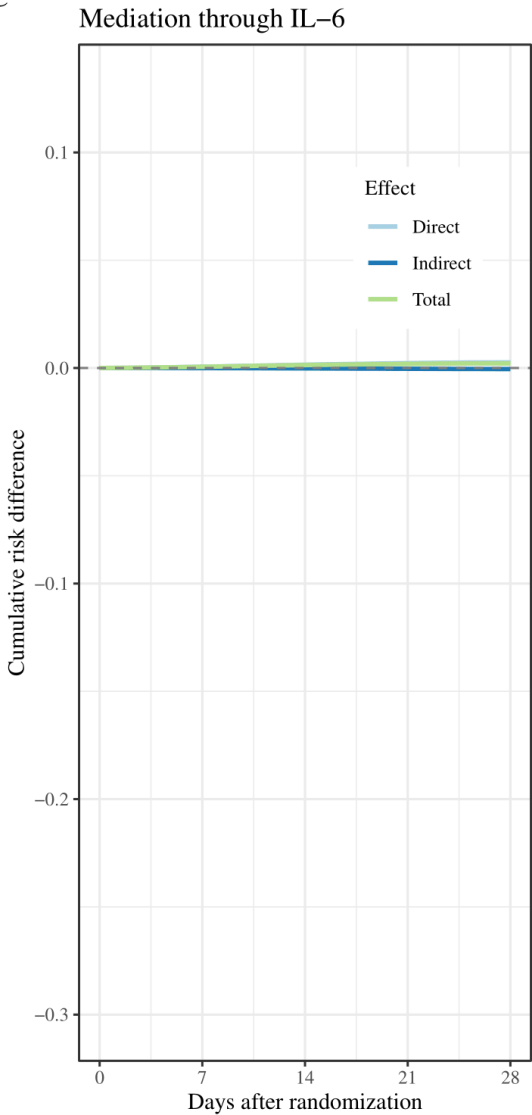
A



B



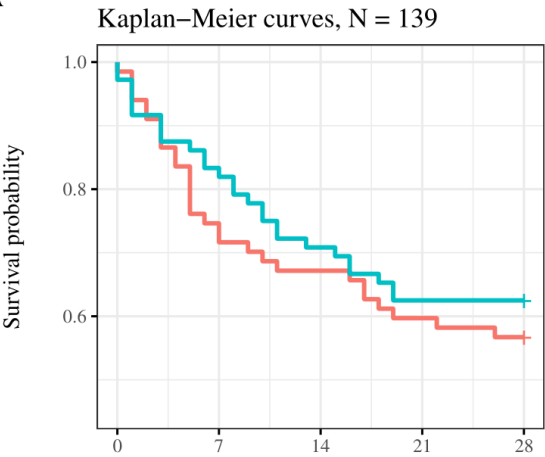
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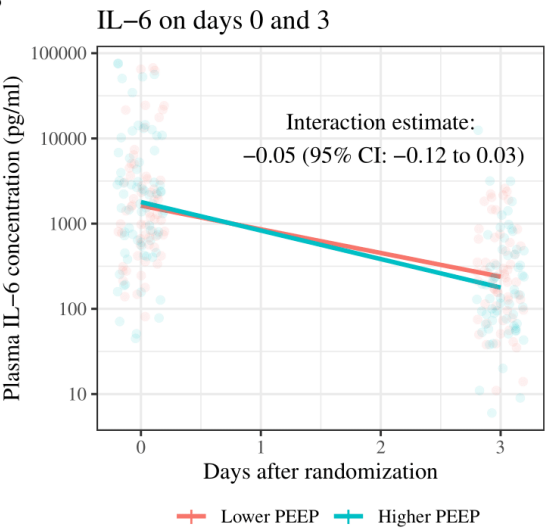
VI: ALVEOLI (hyperinflammatory)

ALVEOLI (hyperinflammatory)

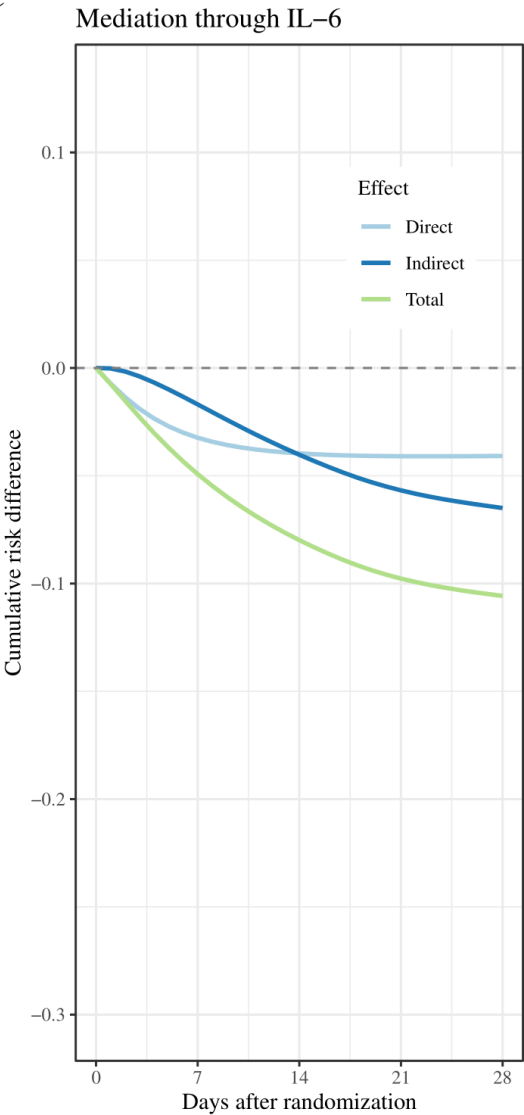
A



B



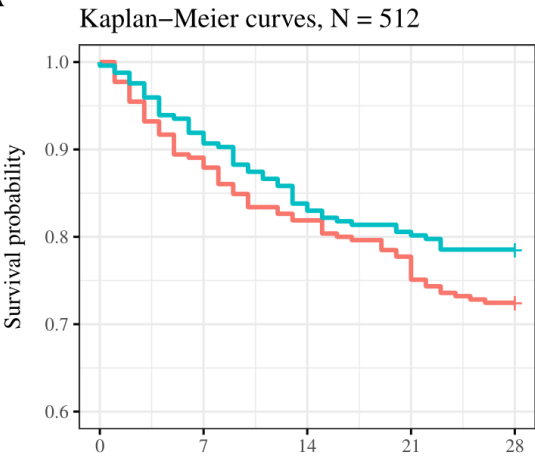
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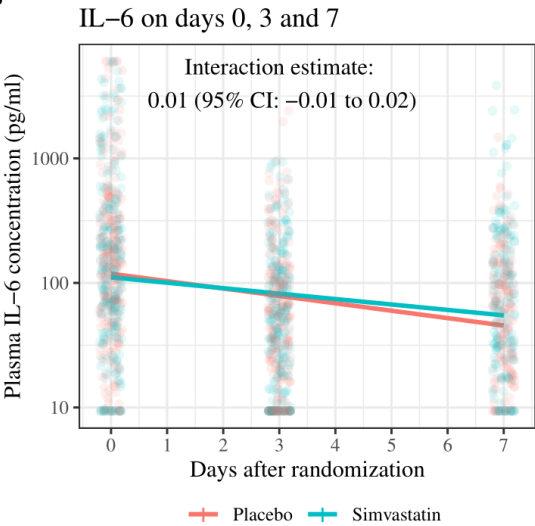
VII: HAPR-2 (all patients)

HARP-2

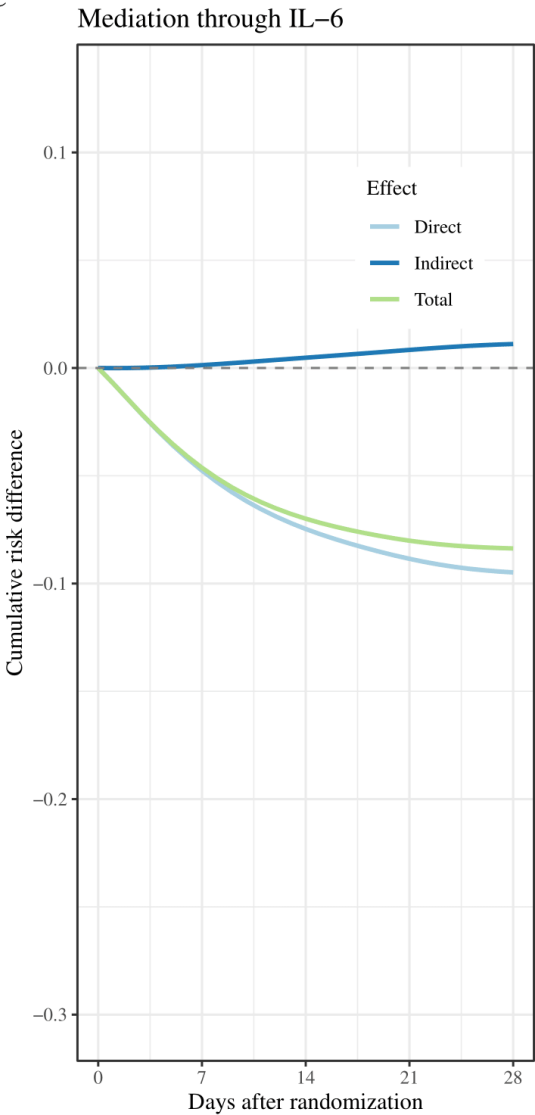
A



B



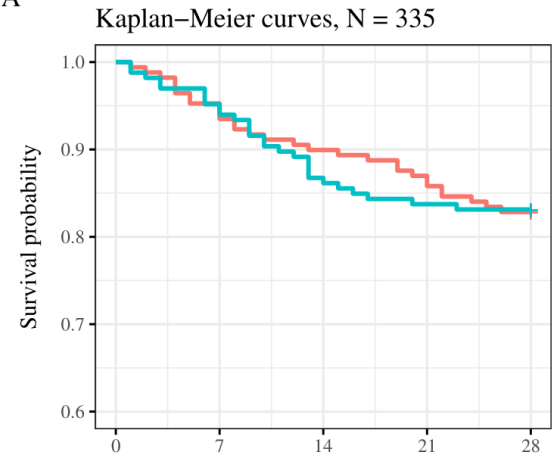
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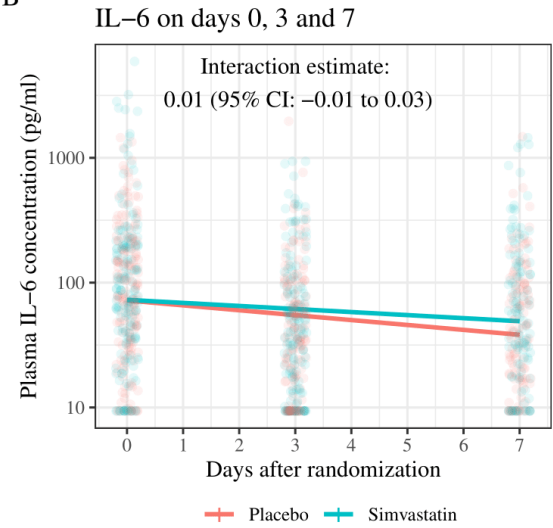
VIII: HARP-2 (hypoinflammatory)

HARP-2 (hypoinflammatory)

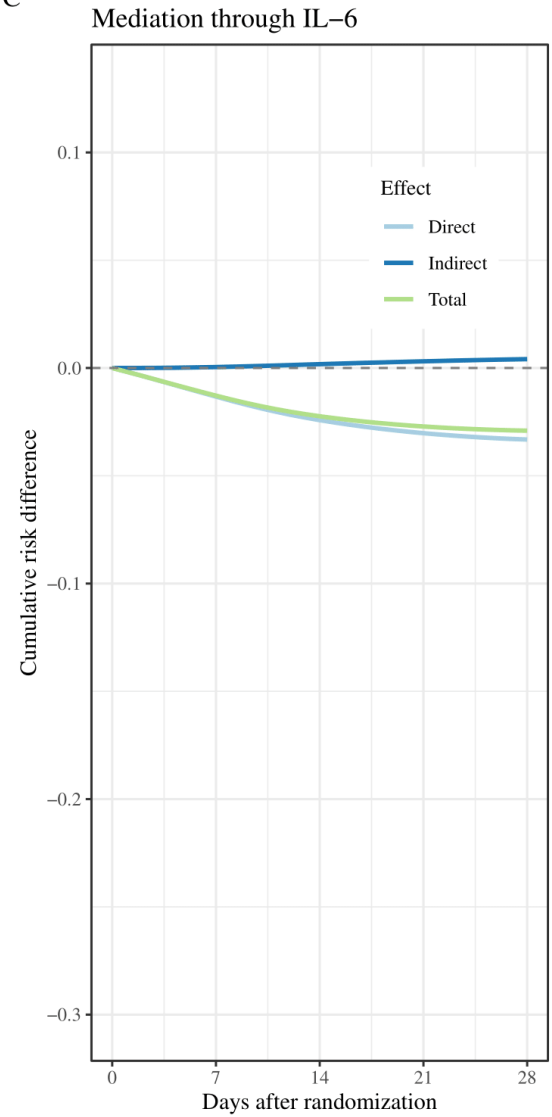
A



B



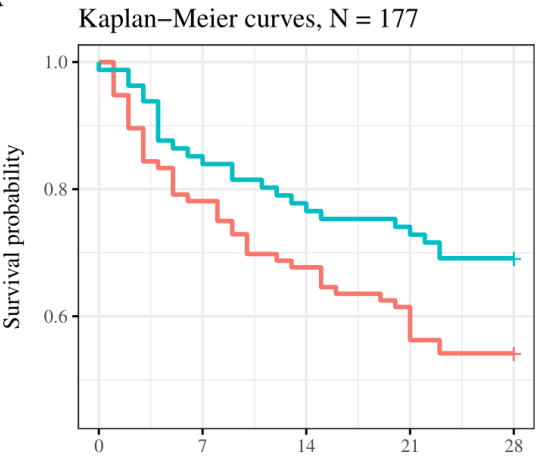
C



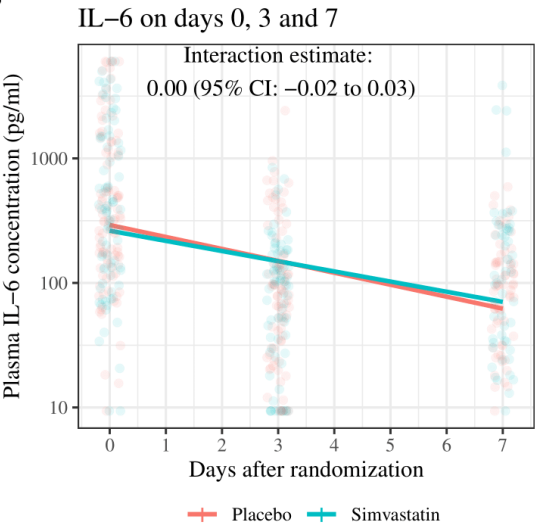
IX: HARP-2 (hyperinflammatory)

HARP2 (hyperinflammatory)

A



B



C

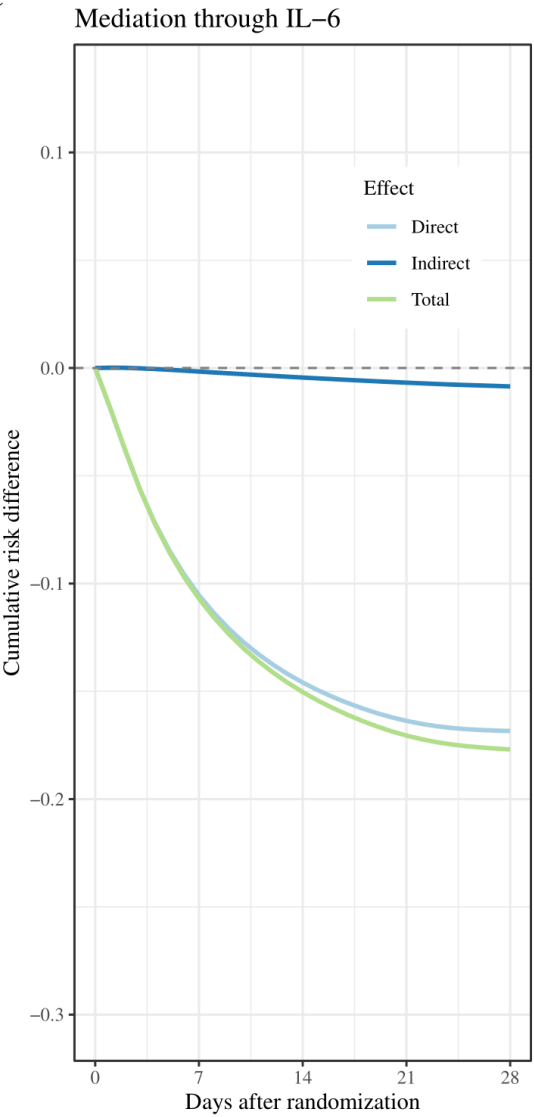
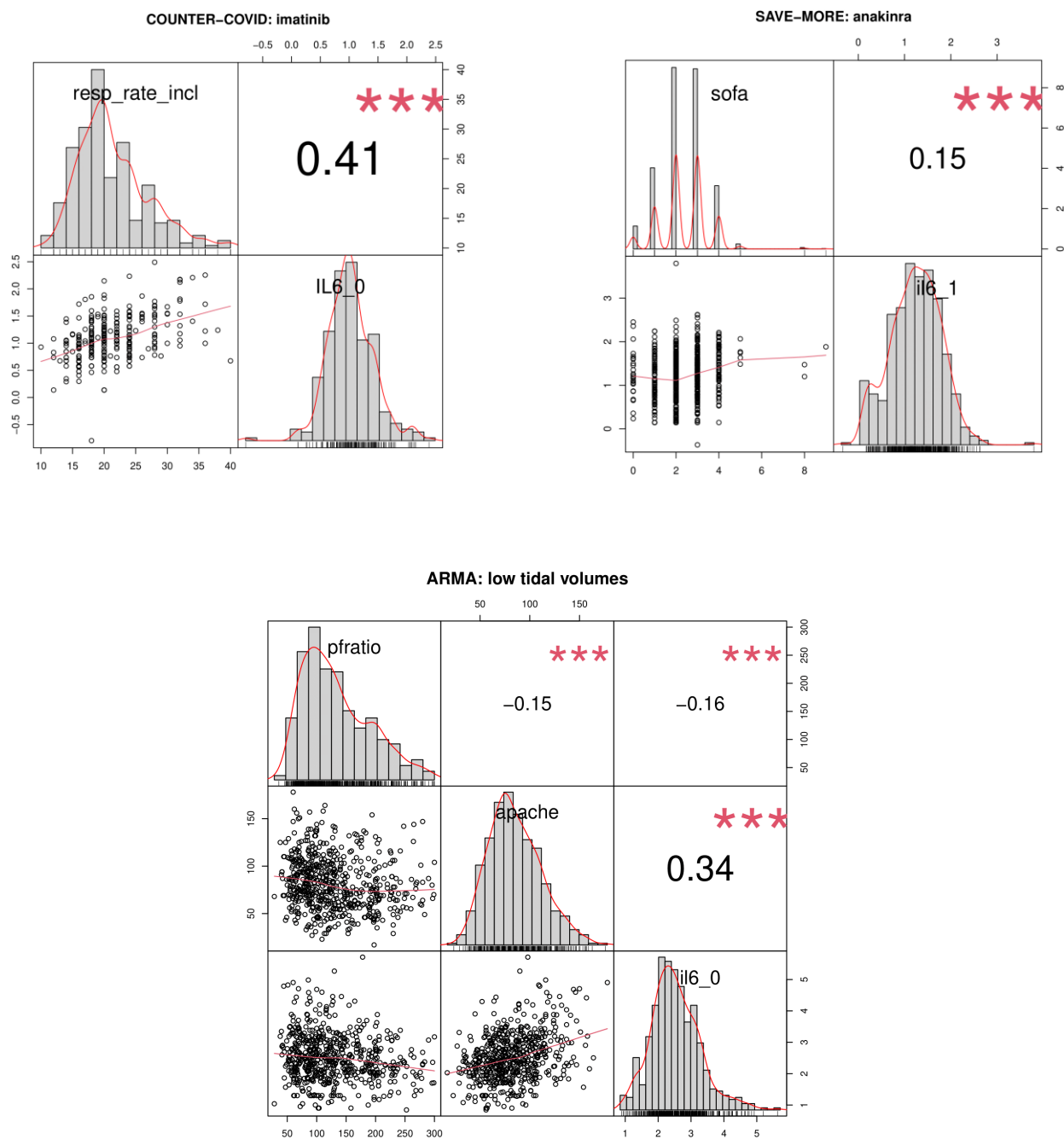
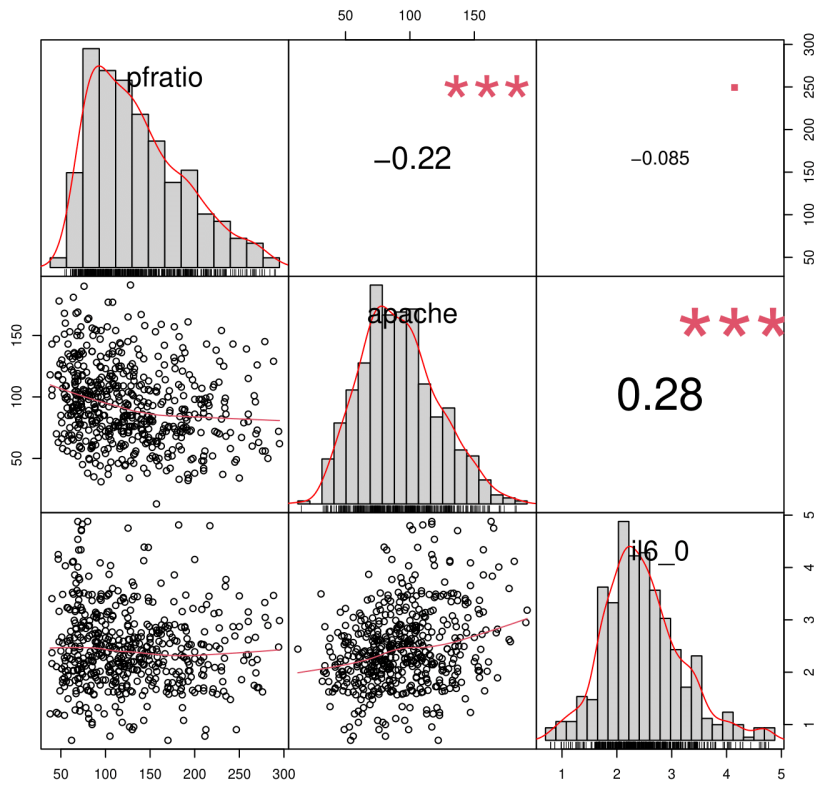


Figure S6: Correlations between IL-6 and indexes of organ failure and/or oxygenation at baseline.



ALVEOLI: high PEEP



HARP-2: simvastatin

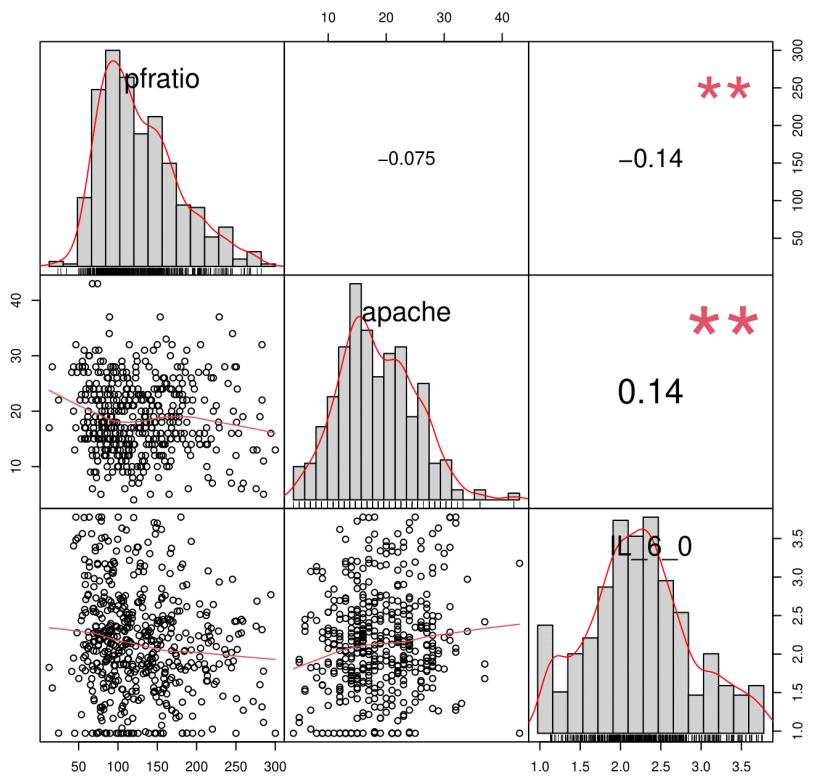


Figure S7: Associations between observed IL-6 changes and estimated mortality risks split by treatment and control groups. Patient-specific mortality risks after day 3 per observed IL-6 reductions within 3 days after randomization. Each point represents one patient. Individual mortality risk was estimated using the joint models for each trial. Points to the left of the vertical dotted lines indicate patients with decreased IL-6 between days 0 and 3. Points to the right of the vertical dotted lines indicate patients with increased IL-6.

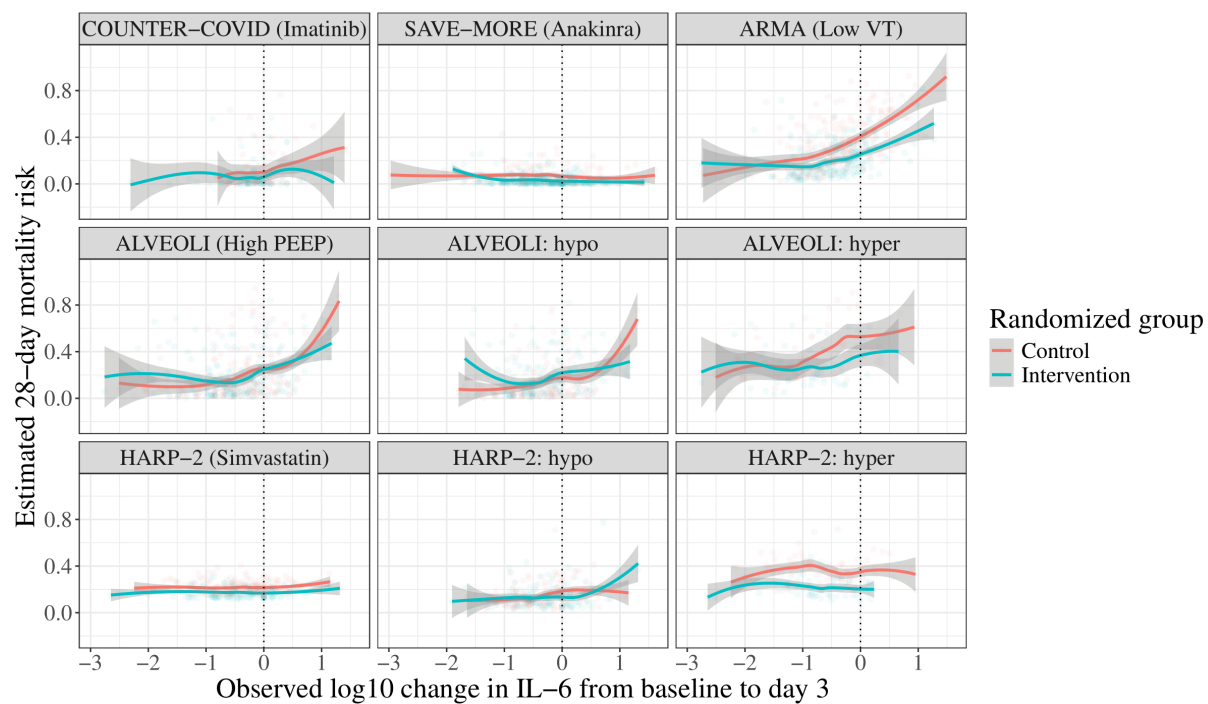
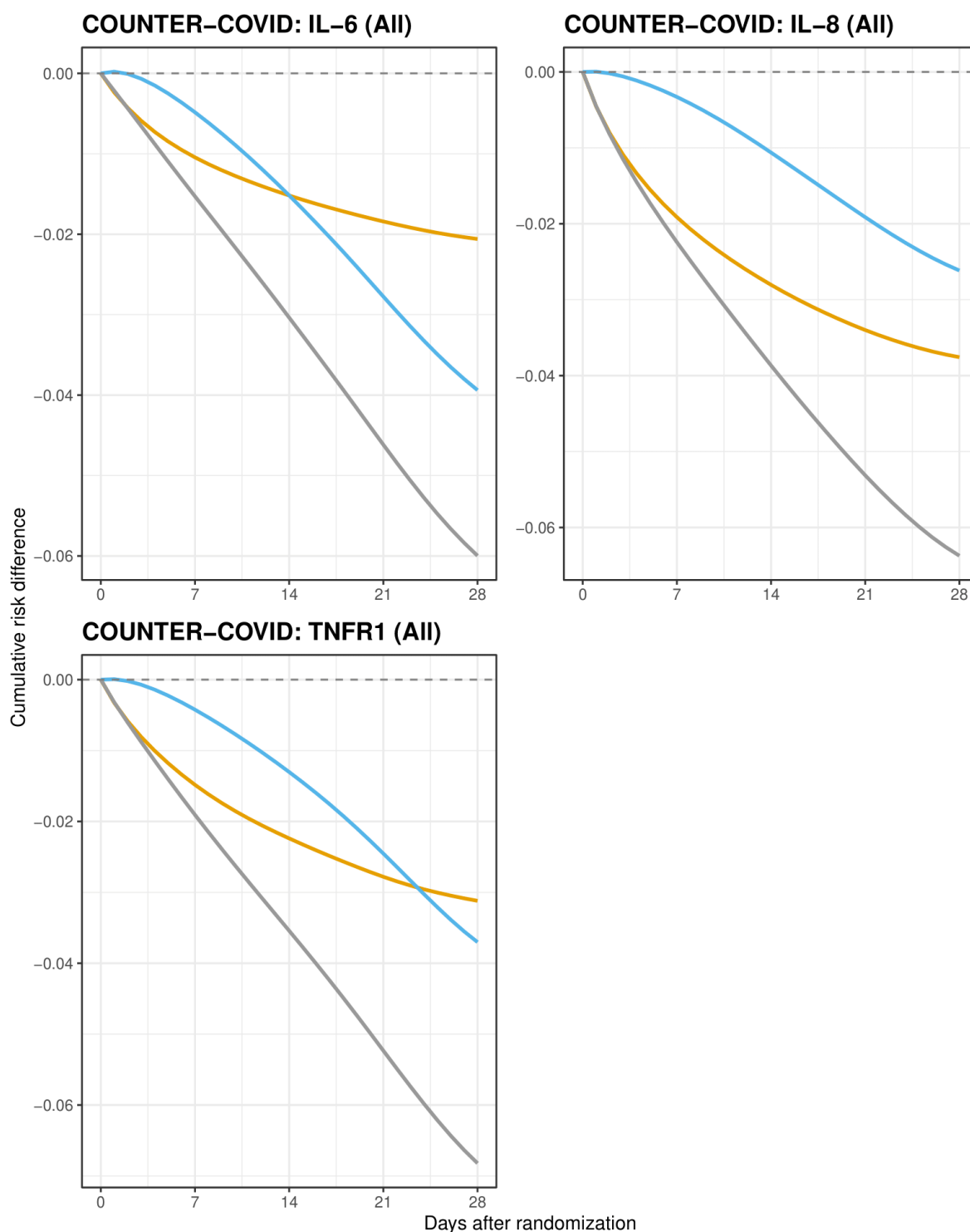


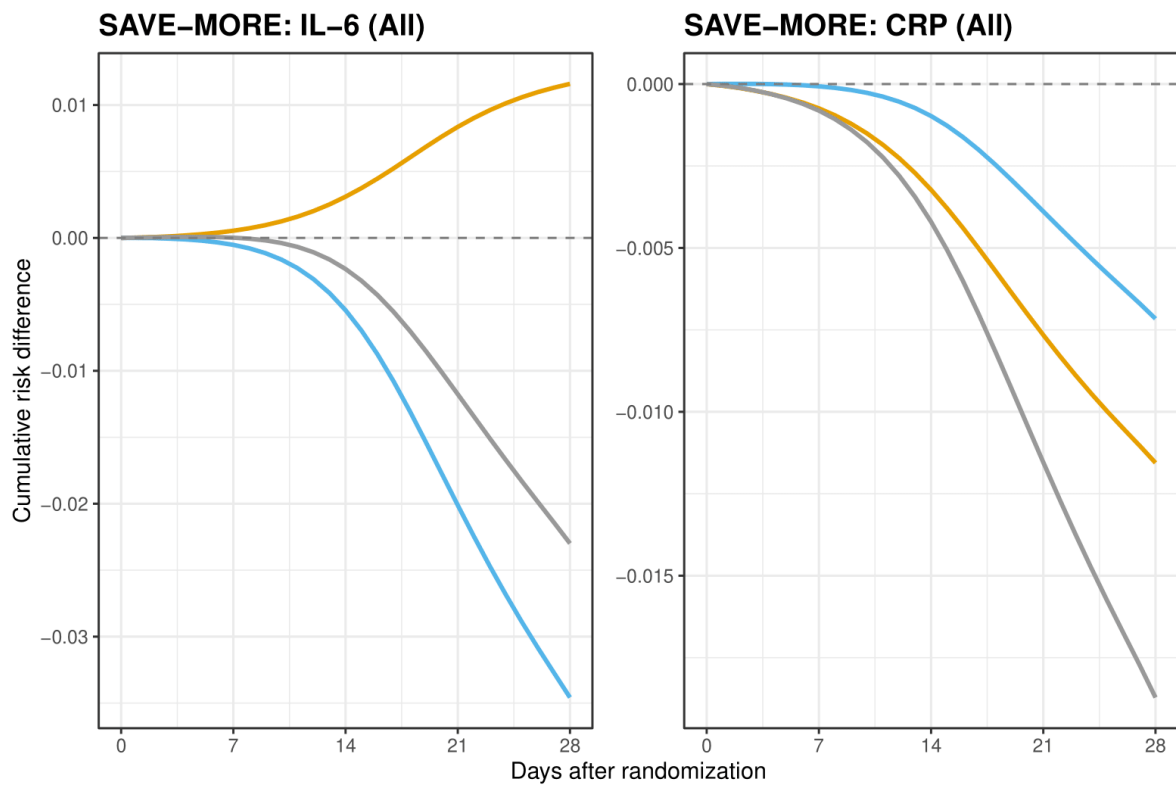
Figure S8: Time-dependent mediation analysis using IL-6, IL-8, TNFR1, and CRP.

Grey lines represent the total risk differences due to intervention, blue lines indirect effects (risk differences due to intervening on the aspect of treatment that affects the biomarker), and orange lines direct effects (risk differences due to intervening on the aspect of treatment that affects the survival outcome, keeping the biomarker at the level it would naturally take on without intervention).

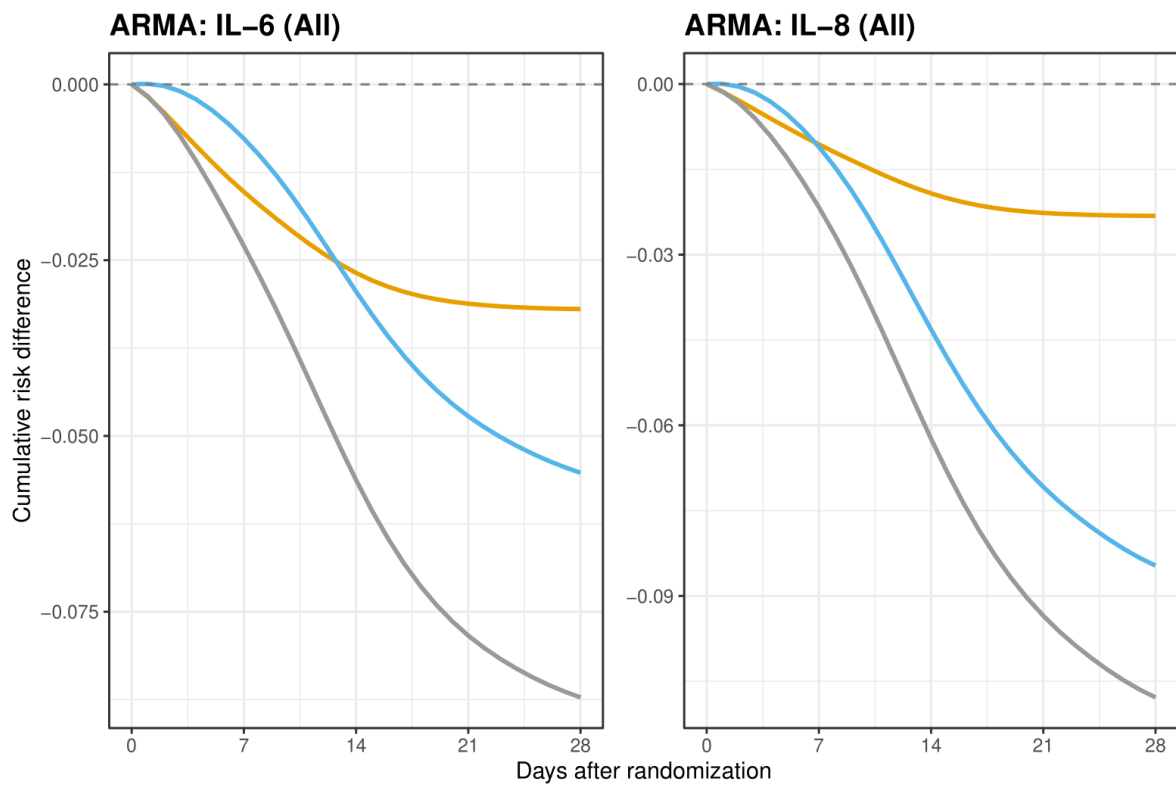
I: COUNTER-COVID



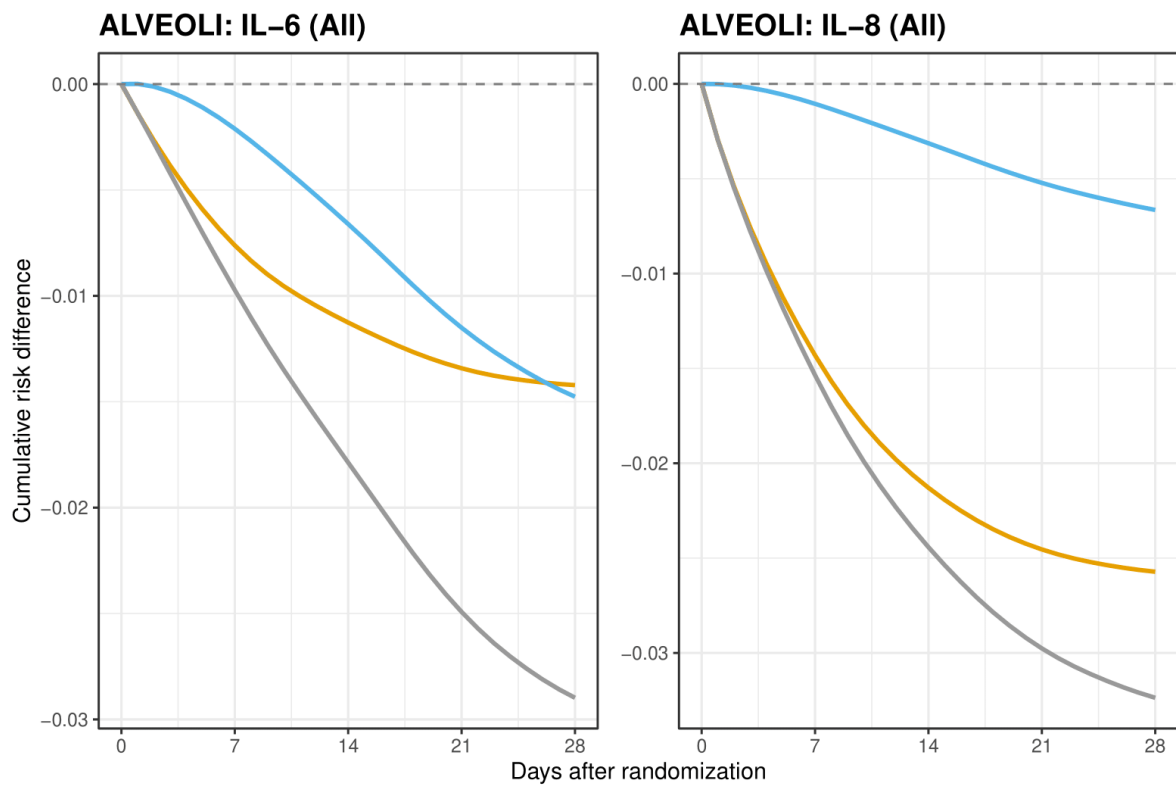
II: SAVE-MORE



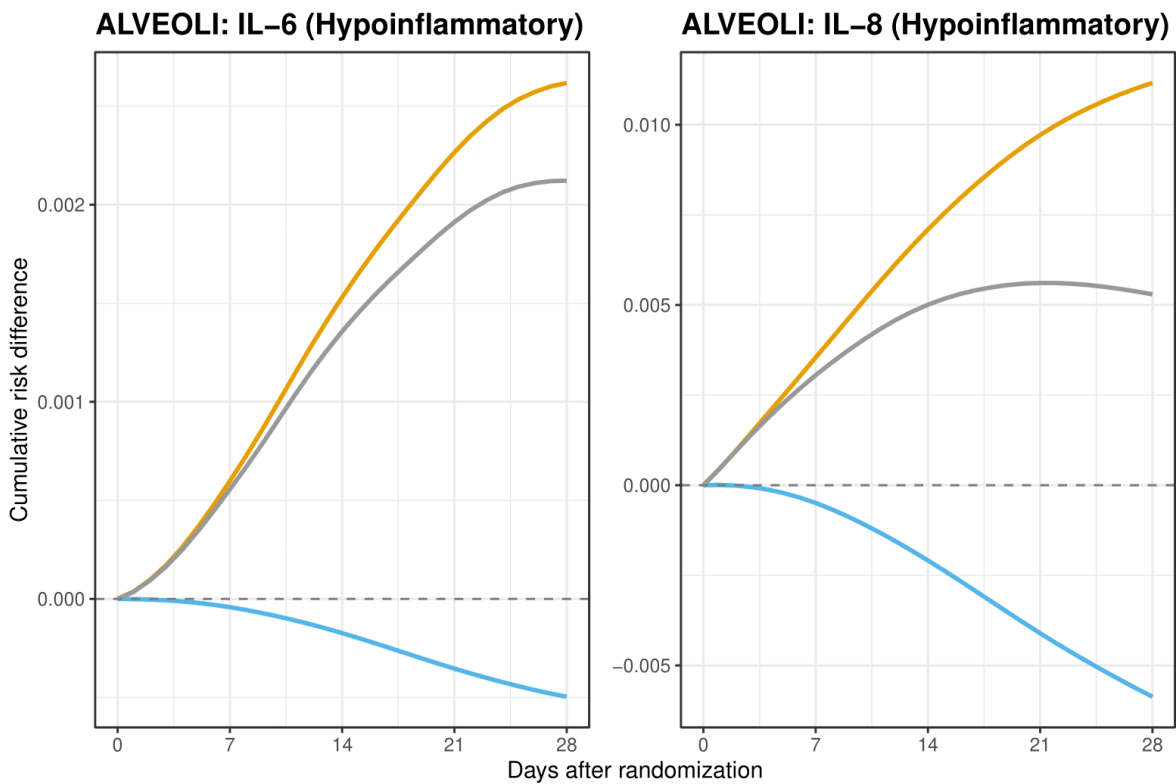
III: ARMA



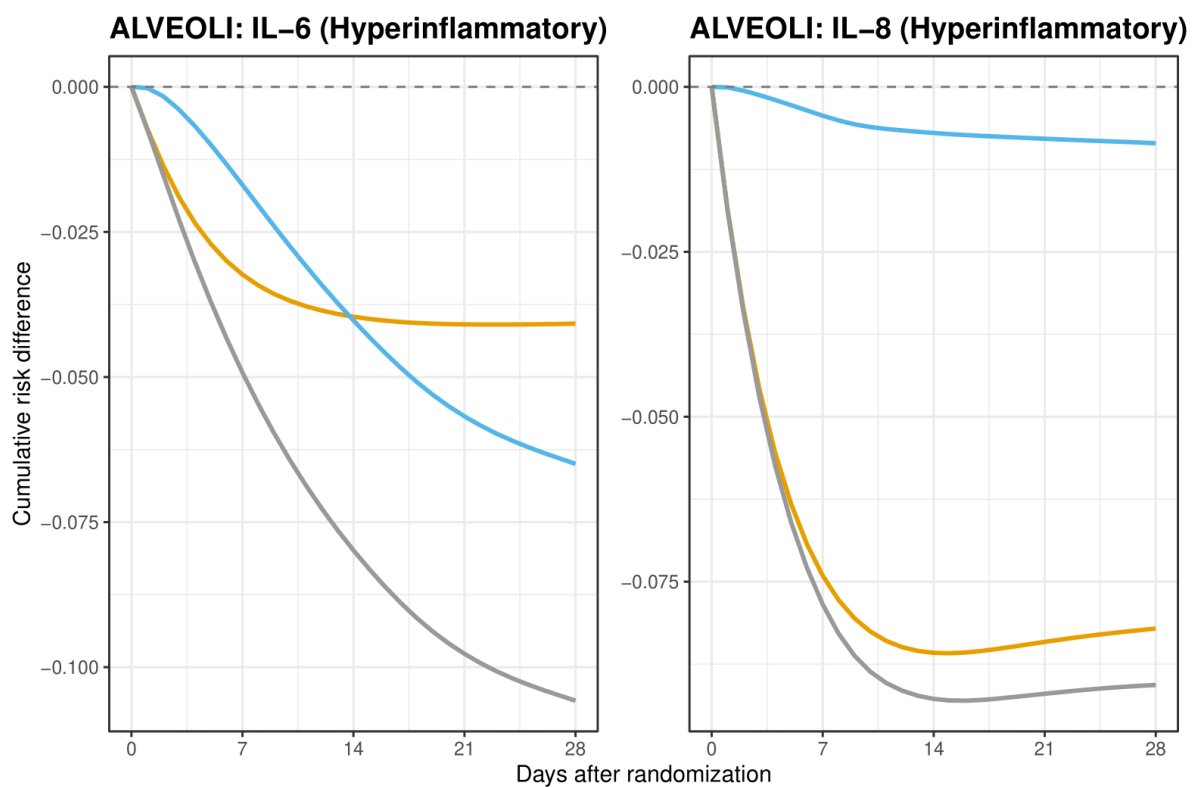
IV: ALVEOLI



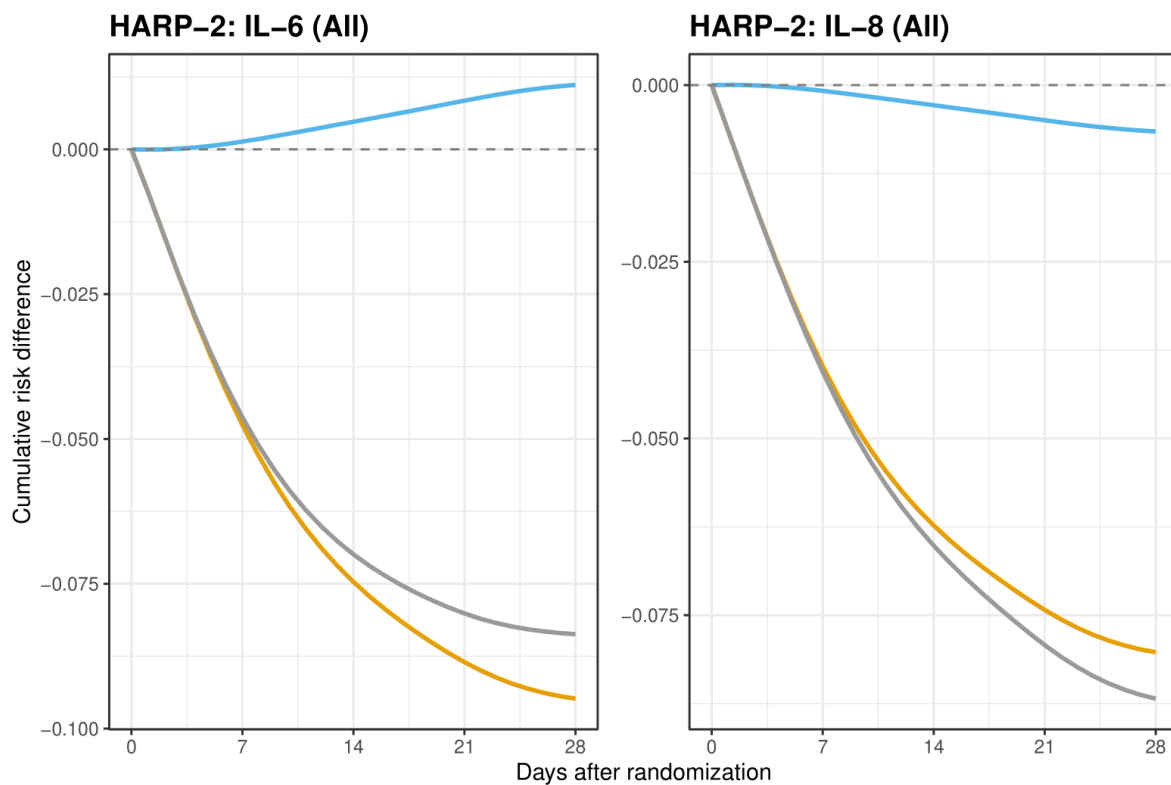
V: ALVEOLI (hypoinflammatory)



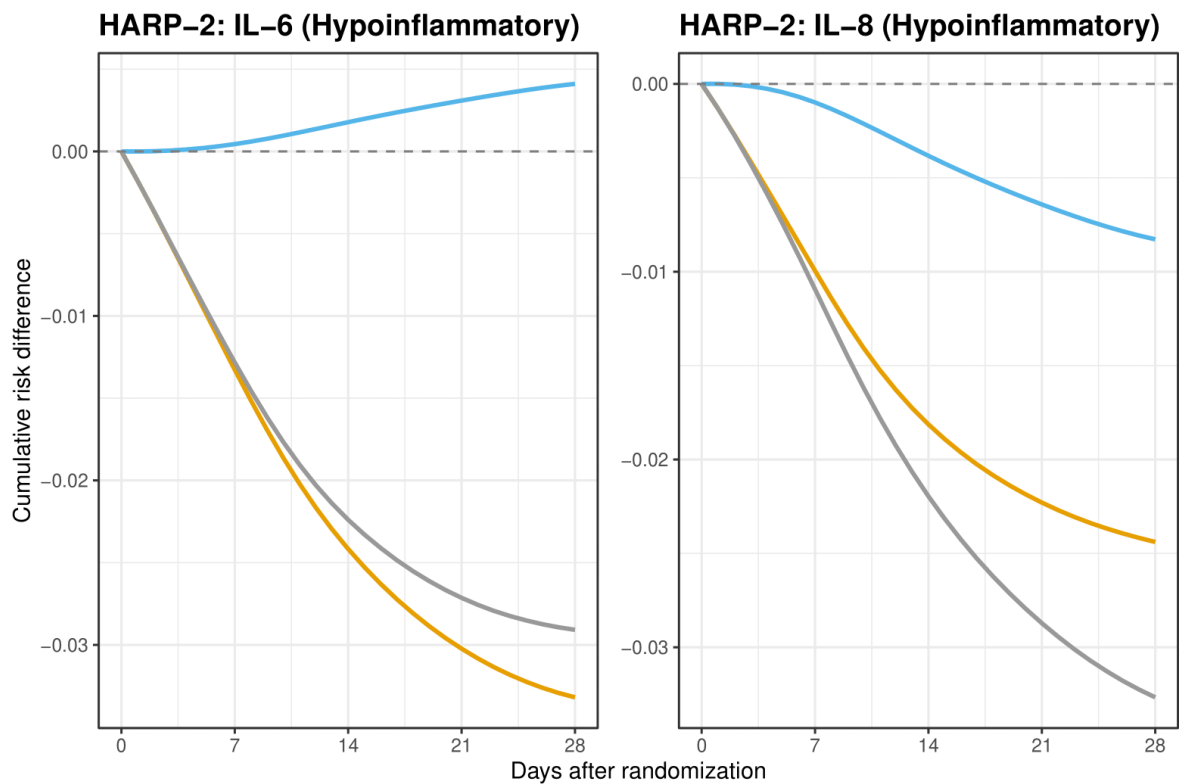
VI: ALVEOLI (hyperinflammatory)



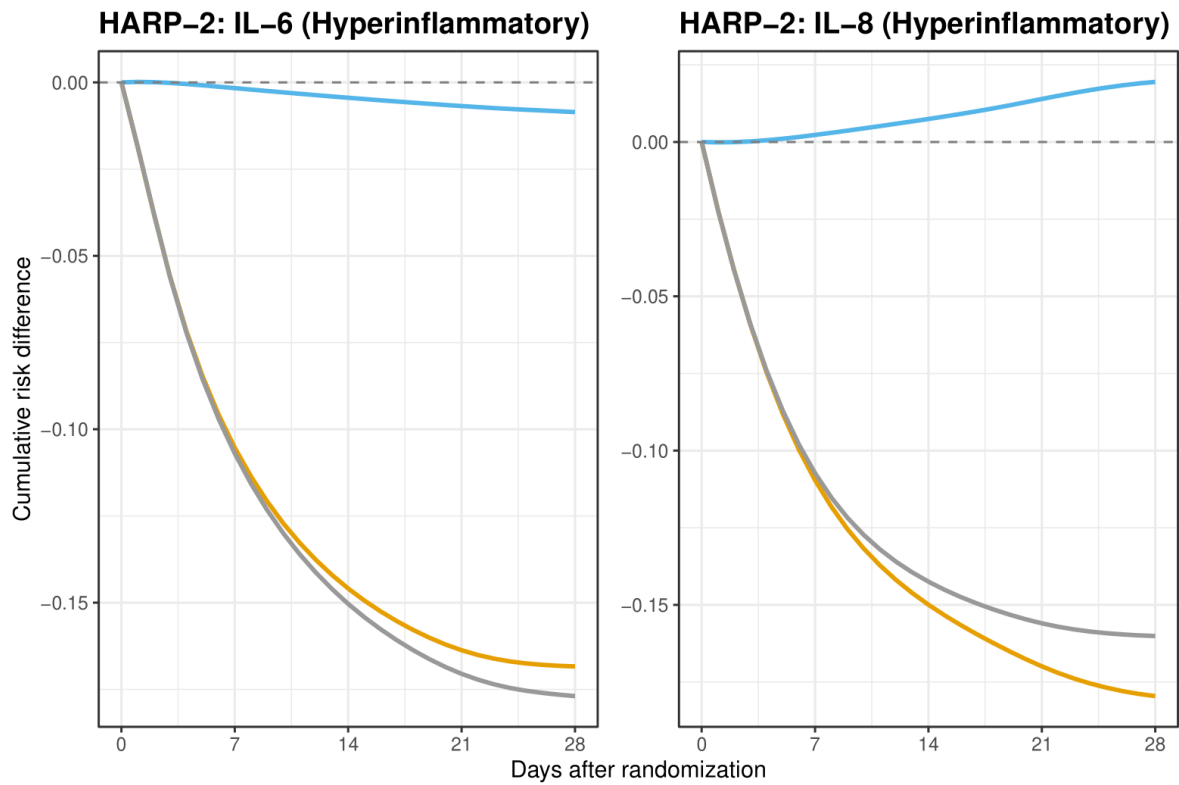
VII: HARP-2 (all patients)



VIII: HARP-2 (hypoinflammatory)



IX: HARP-2 (hyperinflammatory)



TABLES

Table S1: Brief descriptions of the included trials.

COUNTER-COVID:	This study enrolled patients from 13 Dutch hospitals in 2020 and 2021. Patients were included if they had COVID-19, were ≥ 18 years of age, and required supplemental oxygen to maintain peripheral oxygen saturation $>94\%$. Participants were randomized to either treatment with imatinib or a placebo. Plasma biomarker measures were obtained on days 0, 2, 3, and 5.
SAVE-MORE:	This study enrolled patients from 37 hospitals in 2020 and 2021. Patients were included if they had COVID-19, and suPAR $>6 \text{ ng ml}^{-1}$. Participants were randomized to either treatment with anakinra or a placebo. Plasma biomarker measures were obtained on days 0, 3, and 6.
ARMA:	Patient were enrolled from 10 hospitals between 1996 and 1999. Patient were included if they had acute lung injury and ARDS, mechanically ventilated, and ≥ 18 years of age. Participants were randomized to either low tidal volume (6ml per kilogram bodyweight) or traditional/high tidal volume (12ml per kilogram bodyweight). Blood plasma samples of interleukin-6 were obtained on days 0 and 3 by immunoassay.
ALVEOLI:	Patients were enrolled from 23 hospitals between 1999 and 2002. Patients were included if they had acute lung injury and ARDS, mechanically ventilated, and ≥ 18 years of age. Stratified by hospital, participants were randomized to either low or high PEEP. Tidal volume was set at 6ml/kg for all patients. Blood plasma samples of interleukin-6 were obtained on days 0 and 3 by immunoassay.
HARP-2:	This study enrolled patients from 40 hospitals in the UK and Ireland between 2010 and 2014. Patients were included if they had ARDS and were mechanically ventilated. After stratification by site and vasopressor requirement, participants were randomized to treatment with either simvastatin or a placebo. Plasma biomarker measures were obtained on days 0 and 3.

Table S2: Baseline characteristics of trial participants**COUNTER-COVID**

	Imatinib (N=168)	Placebo (N=163)	Overall (N=331)
Age (years)			
Median [Min, Max]	65.0 [28.0, 93.0]	64.0 [9.00, 87.0]	64.0 [9.00, 93.0]
Sex			
Female	42 (25.0%)	55 (33.7%)	97 (29.3%)
Male	126 (75.0%)	108 (66.3%)	234 (70.7%)
BMI (kg/m²)			
Median [Min, Max]	27.2 [17.1, 53.0]	29.8 [18.1, 48.4]	28.3 [17.1, 53.0]
Baseline IL-6 (log10 (pg/mL))			
Mean (SD)	1.03 (0.427)	1.12 (0.412)	1.08 (0.422)

SAVE-MORE

	Placebo (N=188)	Anakinra (N=401)	Overall (N=589)
Age (years)			
Median [Min, Max]	61.5 [23.0, 87.0]	62.0 [22.0, 92.0]	62.0 [22.0, 92.0]
Sex			
Female	81 (43.1%)	166 (41.4%)	247 (41.9%)
Male	107 (56.9%)	235 (58.6%)	342 (58.1%)
BMI (kg/m²)			
Median [Min, Max]	28.6 [17.6, 54.7]	28.4 [17.8, 55.6]	28.4 [17.6, 55.6]
SOFA score			
Median [Min, Max]	2.00 [0, 8.00]	2.00 [0, 9.00]	2.00 [0, 9.00]
Baseline IL-6 (log10 (pg/mL))			
Mean (SD)	1.25 (0.571)	1.20 (0.536)	1.22 (0.547)
	High TV (12 ml/kg) (N=298)	Low TV (6 ml/kg) (N=305)	Overall (N=603)
Age (years)			
Median [Min, Max]	51.5 [18.0, 87.0]	49.0 [19.0, 97.0]	50.0 [18.0, 97.0]
Sex			
Female	128 (43.0%)	124 (40.7%)	252 (41.8%)
Male	170 (57.0%)	181 (59.3%)	351 (58.2%)
BMI (kg/m²)			
Median [Min, Max]	25.5 [14.5, 269]	25.4 [12.0, 58.7]	25.5 [12.0, 269]
APACHE III score			
Median [Min, Max]	82.5 [23.0, 164]	75.0 [17.0, 178]	78.0 [17.0, 178]
PaO2/FiO2 ratio			
Median [Min, Max]	118 [38.0, 299]	116 [28.0, 300]	116 [28.0, 300]
Baseline IL-6 (log10 (pg/mL))			
Mean (SD)	2.54 (0.781)	2.53 (0.730)	2.53 (0.755)

ALVEOLI

	Hypoinflammatory		Hyperinflammatory		Overall	
	Higher PEEP (N=195)	Lower PEEP (N=196)	Higher PEEP (N=72)	Lower PEEP (N=67)	Higher PEEP (N=267)	Lower PEEP (N=263)
Age (years)						
Median [Min, Max]	57.0 [19.0, 88.0]	47.0 [18.0, 86.0]	46.5 [16.0, 86.0]	46.0 [18.0, 86.0]	53.0 [16.0, 88.0]	47.0 [18.0, 86.0]
Sex						
Female	85 (43.6%)	93 (47.4%)	32 (44.4%)	29 (43.3%)	117 (43.8%)	122 (46.4%)
Male	110 (56.4%)	103 (52.6%)	40 (55.6%)	38 (56.7%)	150 (56.2%)	141 (53.6%)
BMI (kg/m²)						
Median [Min, Max]	27.5 [14.0, 55.4]	26.4 [13.8, 58.8]	25.9 [17.7, 52.7]	26.0 [15.7, 52.7]	27.0 [14.0, 55.4]	26.0 [13.8, 58.8]
APACHE III score						
Median [Min, Max]	85.5 [32.0, 160]	81.0 [13.0, 154]	118 [59.0, 191]	111 [50.0, 177]	94.5 [32.0, 191]	89.0 [13.0, 177]
PaO2/FiO2 ratio						
Median [Min, Max]	119 [38.0, 295]	125 [47.0, 294]	91.7 [40.0, 270]	112 [47.0, 260]	112 [38.0, 295]	122 [47.0, 294]
Baseline IL-6 (log10 (pg/mL))						
Mean (SD)	2.17 (0.556)	2.22 (0.560)	3.25 (0.758)	3.21 (0.678)	2.47 (0.781)	2.46 (0.728)

HARP-2

	Hyperinflammatory		Hypoinflammatory		Overall	
	Placebo (N=96)	Simva-statin (N=81)	Placebo (N=169)	Simva-statin (N=166)	Placebo (N=265)	Simva-statin (N=247)
Age (years)						
Median [Min, Max]	62.2 [23.3, 84.6]	60.5 [21.1, 86.0]	51.0 [16.2, 90.1]	49.0 [17.4, 82.9]	54.8 [16.2, 90.1]	53.0 [17.4, 86.0]
Sex						
Female	40 (41.7%)	32 (39.5%)	66 (39.1%)	84 (50.6%)	106 (40.0%)	116 (47.0%)
Male	56 (58.3%)	49 (60.5%)	103 (60.9%)	82 (49.4%)	159 (60.0%)	131 (53.0%)
APACHE II score						
Median [Min, Max]	20.0 [7.00, 32.0]	23.0 [10.0, 43.0]	16.0 [5.00, 37.0]	17.0 [4.00, 43.0]	18.0 [5.00, 37.0]	19.0 [4.00, 43.0]
SOFA score						
Median [Min, Max]	10.0 [5.00, 18.0]	11.0 [6.00, 17.0]	8.00 [3.00, 16.0]	7.00 [3.00, 15.0]	9.00 [3.00, 18.0]	8.00 [3.00, 17.0]
PaO2/FiO2 ratio						
Median [Min, Max]	124 [46.5, 277]	99.7 [43.5, 283]	120 [12.5, 300]	115 [24.5, 285]	123 [12.5, 300]	110 [24.5, 285]
Baseline IL-6 (log10 pg/mL)						
Mean (SD)	2.58 (0.633)	2.59 (0.671)	1.93 (0.536)	1.94 (0.614)	2.16 (0.653)	2.15 (0.702)

Table S3: Joint modeling results for each trial, adjusted for baseline covariates *age*, *sex*, *BMI*, *SOFA/APACHE/respiratory rate*. *Age*, *BMI*, *SOFA*, *respiratory rate*, and *APACHE* were mean-centered before analysis. The longitudinal submodel refers to the mixed effects submodel in the joint model for IL-6 over time with a random intercept for each patient and random slopes for time. The survival submodel refers to the time-to-event model in the joint model for survival to day 28.

COUNTER-COVID: Based on 293 patients with 623 observations of IL-6. Number of events: 32 (10.9%).

	Longitudinal submodel			Survival submodel	
	Mean	SD	95% credibility interval	Hazard ratio	95% credibility interval (HR)
Day	-0.015*	0.012	-0.038 to 0.009	-	-
Day*Imatinib	-0.058*	0.017	-0.091 to -0.025	-	-
IL-6	-	-	-	2.963*	1.344 to 7.591
Imatinib	-	-	-	0.701	0.225 to 2.175
Age	0.007*	0.002	0.003 to 0.011	1.177*	1.116 to 1.247
Sex (male)	0.172*	0.055	0.064 to 0.279	4.989*	1.696 to 17.494
BMI	0.004	0.005	-0.005 to 0.013	1.165*	1.062 to 1.285
RR	0.003*	0.001	0.001 to 0.005	1.069*	1.022 to 1.113
sigma	0.307	0.015	0.28 to 0.337	-	-

SAVE-MORE: Based on 586 patients with 1659 observations of IL-6. Number of events: 26 (4.4%).

	Longitudinal submodel			Survival submodel	
	Mean	SD	95% credibility interval	Hazard ratio	95% credibility interval (HR)
Day	-0.042*	0.007	-0.056 to -0.029	-	-
Day*Anakinra	-0.025*	0.007	-0.039 to -0.01	-	-
IL-6	-	-	-	21.832*	5.279 to 126.007
Anakinra	-	-	-	0.925	0.205 to 4.592
Age	0.005*	0.001	0.002 to 0.007	1.123*	1.056 to 1.201
Sex (male)	0.093*	0.036	0.022 to 0.164	2.985	0.839 to 12.071
BMI	-0.003	0.003	-0.009 to 0.003	0.907	0.767 to 1.045
SOFA	0.07*	0.016	0.039 to 0.102	1.381	0.915 to 1.997
sigma	0.439	0.011	0.417 to 0.461	-	-

ARMA: Based on 550 patients with 1020 observations of IL-6. Number of events: 167 (30.4%).

	Longitudinal submodel			Survival submodel	
	Mean	SD	95% credibility interval	Hazard ratio	95% credibility interval (HR)
Day	-0.110*	0.012	-0.134 to -0.087	-	-
Day*low TV	-0.038*	0.015	-0.067 to -0.009	-	-
IL-6	-	-	-	2.412*	1.313 to 4.511
low TV	-	-	-	0.860	0.531 to 1.447
Age	-0.004*	0.002	-0.007 to -0.001	1.028*	1.016 to 1.042
Sex (male)	-0.022	0.053	-0.126 to 0.082	1.018	0.715 to 1.462
BMI	-0.009*	0.004	-0.016 to -0.001	0.987	0.958 to 1.018
APACHE III	0.009*	0.001	0.007 to 0.011	1.015*	1.007 to 1.022
sigma	0.331	0.056	0.191 to 0.396	-	-

ALVEOLI: Based on 482 patients with 897 observations of IL-6. Number of events: 113 (23.4%).

	Longitudinal submodel			Survival submodel	
	Mean	SD	95% credibility interval	Hazard ratio	95% credibility interval (HR)
Day	-0.144*	0.013	-0.170 to -0.119	-	-
Day*high PEEP	-0.014	0.016	-0.046 to 0.015	-	-
IL-6	-	-	-	2.709*	1.171 to 7.402
high PEEP	-	-	-	0.938	0.536 to 1.673
Age	0.001	0.002	-0.002 to 0.004	1.038*	1.024 to 1.053
Sex (male)	0.064	0.056	-0.045 to 0.174	0.969	0.628 to 1.482
BMI	0.000	0.004	-0.008 to 0.008	1.020	0.988 to 1.055
APACHE III	0.005*	0.001	0.003 to 0.007	1.028*	1.02 to 1.035
sigma	0.385	0.069	0.117 to 0.456	-	-

ALVEOLI (hyperinflammatory): Based on 126 patients with 229 observations of IL-6. Number of events: 53 (42.1%).

	Longitudinal submodel			Survival submodel	
	Mean	SD	95% credibility interval	Hazard ratio	95% credibility interval (HR)
Day	-0.266*	0.031	-0.327 to -0.206	-	-
Day*high PEEP	-0.047	0.037	-0.120 to 0.026	-	-
IL-6	-	-	-	1.953*	1.228 to 3.809
high PEEP	-	-	-	0.860	0.387 to 1.886
Age	0.003	0.003	-0.003 to 0.009	1.033*	1.012 to 1.054
Sex (male)	0.051	0.105	-0.155 to 0.256	0.936	0.506 to 1.748
BMI	-0.008	0.007	-0.023 to 0.006	1.022	0.974 to 1.073
APACHE III	0.000	0.002	-0.004 to 0.003	1.020*	1.009 to 1.033
sigma	0.372	0.088	0.175 to 0.511	-	-

ALVEOLI (hypoinflammatory): Based on 356 patients with 668 observations of IL-6. Number of events: 60 (16.9%).

	Longitudinal submodel			Survival submodel	
	Mean	SD	95% credibility interval	Hazard ratio	95% credibility interval (HR)
Day	-0.103*	0.014	-0.13 to -0.076	-	-
Day*high PEEP	-0.005	0.018	-0.041 to 0.03	-	-
IL-6	-	-	-	1.283*	1.028 to 1.860
high PEEP	-	-	-	0.981	0.503 to 1.932
Age	0.002	0.001	-0.001 to 0.005	1.048*	1.029 to 1.068
Sex (male)	0.062	0.051	-0.037 to 0.164	1.087	0.643 to 1.868
BMI	0.003	0.004	-0.005 to 0.01	1.015	0.972 to 1.058
APACHE III	0.001	0.001	-0.001 to 0.003	1.029*	1.019 to 1.039
sigma	0.263	0.045	0.168 to 0.338	-	-

HARP-2: Based on 453 patients with 863 observations of IL-6. Number of events: 110 (24.3%).

	Longitudinal submodel			Survival submodel	
	Mean	SD	95% credibility interval	Hazard ratio	95% credibility interval (HR)
Day	-0.055*	0.007	-0.068 to -0.042	-	-
Day*Simvastatin	0.006	0.008	-0.009 to 0.022	-	-
IL-6	-	-	-	4.965*	2.017 to 13.716
Simvastatin	-	-	-	0.581	0.332 to 0.996
Age	0.002	0.001	-0.001 to 0.005	1.015*	1.001 to 1.03
Sex (male)	0.111	0.045	0.022 to 0.2	0.812	0.537 to 1.216
APACHE III	0.011*	0.004	0.004 to 0.018	1.055*	1.021 to 1.089
sigma	0.459	0.015	0.429 to 0.488	-	-

HARP-2 (hyperinflammatory): Based on 161 patients with 302 observations of IL-6. Number of events: 63 (39.1%).

	Longitudinal submodel			Survival submodel	
	Mean	SD	95% credibility interval	Hazard ratio	95% credibility interval (HR)
Day	-0.088*	0.012	-0.11 to -0.065	-	-
Day*Simvastatin	0.002	0.013	-0.021 to 0.028	-	-
IL-6	-	-	-	9.024*	1.543 to 78.195
Simvastatin	-	-	-	0.508	0.216 to 1.165
Age	0.002	0.003	-0.003 to 0.007	1.002	0.979 to 1.026
Sex (male)	0.065	0.073	-0.078 to 0.208	0.781	0.417 to 1.416
APACHE III	-0.005	0.006	-0.017 to 0.007	1.048	0.995 to 1.111
sigma	0.538	0.026	0.49 to 0.59	-	-

HARP-2 (hypoinflammatory): Based on 292 patients with 561 observations of IL-6. Number of events: 47 (16.1%).

	Longitudinal submodel			Survival submodel	
	Mean	SD	95% credibility interval	Hazard ratio	95% credibility interval (HR)
Day	-0.041*	0.008	-0.057 to -0.025	-	-
Day*Simvastatin	0.011	0.01	-0.009 to 0.031	-	-
IL-6	-	-	-	1.512	0.710 to 3.866
Simvastatin	-	-	-	0.796	0.367 to 1.718
Age	-0.002	0.002	-0.005 to 0.002	1.020	0.999 to 1.042
Sex (male)	0.129*	0.054	0.024 to 0.235	1.069	0.582 to 1.992
APACHE III	0.011*	0.004	0.002 to 0.019	1.094*	1.046 to 1.144
sigma	0.379	0.018	0.345 to 0.416	-	-

Table S4: Comparison of baseline characteristics and mortality between patients with no plasma biomarker measurements (i.e., no IL-6 observations) and patients with at least one measurement.

COUNTER-COVID

	Missing all IL-6 observation	
	FALSE	TRUE
	N = 331	N = 55
Deaths to day 28	37 (11.2%)	5 (9.1%)
Sex		
Male	234 (70.7%)	31 (56.4%)
Age (years)		
Median [IQR]	64.00 [56.00, 73.50]	62.00 [57.00, 70.00]
BMI		
Median [IQR]	28.33 [25.44, 32.27]	28.72 [25.88, 31.90]
Respiratory rate (bpm)		
Median [IQR]	20.00 [18.00, 24.00]	20.00 [17.00, 24.00]

SAVE-MORE

	Missing all IL-6 observation	
	FALSE	TRUE
	N = 589	N = 5
Deaths to day 28		
n (%)	26 (4.4%)	0
NA	1 (0.2%)	0
Sex		
Male	342 (58.1%)	2 (40.0%)
Age (years)		
Median [IQR]	62.00 [54.00, 71.00]	55.00 [51.00, 56.00]
BMI		
Median [IQR]	28.40 [25.88, 32.10]	28.58 [26.12, 29.38]
SOFA		
Median [IQR]	2.00 [2.00, 3.00]	3.00 [2.00, 4.00]

ARMA

	Missing all IL-6 observation	
	FALSE	TRUE
	N = 606	N = 61
Deaths to day 28		
n (%)	183 (30.2%)	21 (65.6%)
NA	3 (0.5%)	0
Sex		
Male	351 (57.9%)	41 (67.2%)
Age (years)		
Median [IQR]	50.50 [37.00, 66.00]	52.00 [40.00, 66.00]
BMI		
Median [IQR]	25.37 [22.28, 29.73]	26.79 [23.71, 29.61]
APACHE II		
Median [IQR]	78.50 [63.00, 100.00]	78.00 [60.00, 99.00]

ALVEOLI

	Missing all IL-6 observation	
	FALSE	TRUE
	N = 530	N = 18
Deaths to day 28		
n (%)	118 (22.3%)	5 (27.8%)
NA	3 (0.5%)	0
Sex		
Male	291 (54.9%)	11 (61.1%)
Age (years)		
Median [IQR]	50.00 [39.00, 65.00]	50.00 [38.25, 54.75]
BMI		
Median [IQR]	26.64 [22.37, 30.47]	
APACHE II		
Median [IQR]	92.00 [71.75, 114.00]	82.50 [63.75, 106.75]

HARP-2

	Missing all IL-6 observation	
	FALSE	TRUE
	N = 512	N = 28
Deaths to day 28		
n (%)	126 (24.6%)	6 (21.4%)
NA	0	1 (3.6%)
Sex		
Male	290 (56.6%)	17 (60.7%)
Age (years)		
Median [IQR]	54.26 [42.41, 66.22]	53.70 [37.19, 65.58]
APACHE III		
Median [IQR]	18.00 [14.00, 23.75]	16.00 [13.50, 21.00]"

Table S5: Missingness of IL-6 observations and number of deaths in IL-6 observation period per trial. *N at risk* excludes patients missing all IL6 measurements and decreases over time as patients die.

COUNTER-COVID

55 patients with no biomarker measurements and 0 with no mortality data were excluded..

Time Point	N at Risk	Available	Missing	% Missing	Cumulative Deaths
Missing all IL-6 measurements	386	331	55	14.2	0
Day 0 (Baseline)	331	295	36	10.9	0
Day 2	329	226	103	31.3	2
Day 3	326	30	296	90.8	5
Day 5	317	136	181	57.1	14

SAVEMORE

5 patients with no biomarker measurements and 1 with no mortality data were excluded.

Time Point	N at Risk	Available	Missing	% Missing	Cumulative Deaths
Missing all IL-6 measurements	594	589	5	0.8	0
Day 1 (Baseline)	588	575	13	2.2	0
Day 4	588	556	32	5.4	0
Day 7	587	533	54	9.2	1

ARMA

61 patients with no biomarker measurements and 3 with no mortality data were excluded.

Time Point	N at Risk	Available	Missing	% Missing	Cumulative Deaths
Missing all IL-6 measurements	667	606	61	9.1	0
Day 0 (Baseline)	603	593	10	1.7	0
Day 3	585	541	44	7.5	18

ALVEOLI

18 patients with no biomarker measurements and 0 with no mortality data were excluded.

Time Point	N at Risk	Available	Missing	% Missing	Cumulative Deaths
Missing all IL-6 measurements	548	530	18	3.3	0
Day 0 (Baseline)	530	520	10	1.9	0
Day 3	512	468	44	8.6	18

HARP-2

28 patients with no measurements and 1 with no mortality data were excluded.

Time Point	N at Risk	Available	Missing	% Missing	Cumulative Deaths
Missing all IL-6 measurements	540	512	28	5.2	0
Day 0 (Baseline)	512	510	2	0.4	0
Day 3	494	461	33	6.7	18
Day 7	463	341	122	26.3	49

Supplement B

February 2026

This document includes detailed information for the mediation analysis approach used in "Interleukin-6 is an effective mediator of therapeutic efficacy in acute respiratory distress syndrome: an individual patient data meta-analysis of RCTs".

1 Present data

The full analysis included five data sets from five randomized controlled trials which evaluated the effectiveness of different interventions in critically ill ARDS and COVID-19 patients. The aim of the study was to investigate whether the cytokine interleukin-6 (IL-6) was a mediator of treatment effects on patient survival.

For all trials, plasma IL-6 was measured on at least two time points during the first week of ICU admission. Patient survival times were available for each trial. For two trials, patients were classified into either a hypo-inflammatory and hyper-inflammatory phenotype, based on previous evidence of treatment effectiveness in one of these subgroups. We conducted nine separate mediation analyses to evaluate whether IL-6 was a mediator of the interventions' effects on patient survival. Here, we provide details for the mediation analysis approach using one data set, namely data from the ARMA trial, which evaluated the effectiveness of lower versus higher tidal volumes during mechanical ventilation.

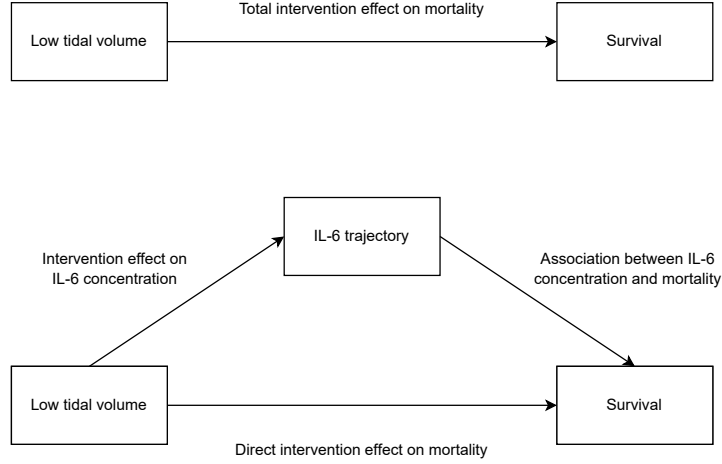
2 Mediation framework

Mediation analysis decomposes the effect of an exposure on an outcome into an indirect effect through an intermediate variable and a direct effect that is independent of the mediator. In the present example, the total effect is the overall effect of low tidal volume on patient survival. The direct effect is the effect of low tidal volume on survival independent of IL-6 concentration. The indirect effect is the effect of low tidal volume on survival mediated through IL-6. The exposure, mediator and outcome are:

- **Exposure:** Low tidal volume (*Binary variable: exposure versus control (high tidal volume)*).
- **Mediator:** Plasma IL-6 concentration, measured repeatedly over time within the first 7 days of intensive care unit admission. (*Repeatedly measured, continuous variable*).
- **Outcome:** Survival time for 28-day endpoint. (*Time-to-event variable*).

In order to obtain estimates of the total, direct, and indirect effects, we need to specify and model of the mediator and a model of the outcome. This will be done in the next section, using a joint modeling approach for longitudinal and survival data.

2.1 Visual representation of the underlying mechanism



3 Joint Modeling Approach

Joint models allow the simultaneous analysis of the longitudinal trajectories of biological markers and the survival data. This method accounts for the potential correlation between longitudinal biomarker measurements and the time-to-event outcome, offering the scope to assess mediation effects. Detailed explanations of joint models can be found elsewhere. [1] Here, we define two sub-models: (1) a longitudinal model for the mediator IL-6, and (2) a survival model for the time-to-event outcome.

3.1 Mediator model: longitudinal submodel

A mixed effects model is used to model the mediator IL-6. The model is defined as

$$\begin{aligned}
 y_i(t) &= \beta_0 + \beta_1 \text{lowTV} \times \text{time} + \beta_2 \text{time} + b_{i1} + b_{i2} \text{time} + \epsilon_i(t) \\
 \epsilon_i(t) &\sim N(0, \sigma^2) \\
 b_{i1}, b_{i2} &\sim N(0, D),
 \end{aligned} \tag{1}$$

where the observed value on the biomarker y of an individual i at time t is a function of fixed and random effects and some time- and person-dependent measurement error $\epsilon_i(t)$. $\epsilon_i(t)$ is assumed to be normally distributed with a mean of 0 and variance σ^2 . The fixed effects include the intercept β_0 , the interaction effect between low tidal volume and time β_1 , and the effect of time β_2 . The random effects are the person-specific random intercept b_{i1} , and the random slope for time b_{i2} . Note that we do not include a fixed effect for the exposure, thereby constraining the mean IL-6 concentrations to be equal at baseline between the exposure (low tidal volume) and the control group (high tidal volume). This constraint is on line with recommendations, e.g., in Fitzmaurice et al. (2011).

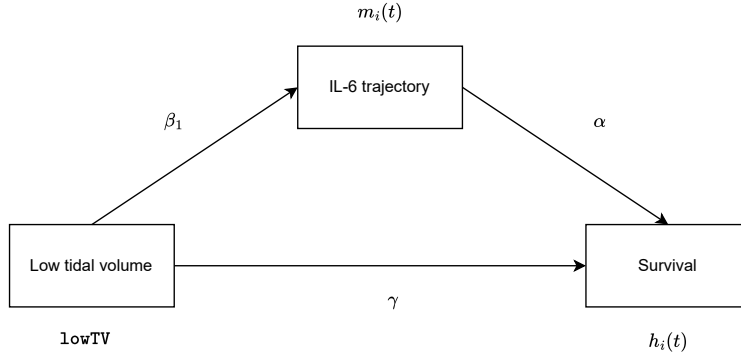
3.2 Outcome model: survival submodel

A time-to-event model is used for patients' survival outcomes. The hazard of an individual i at time t can be expressed as

$$h_i(t) = h_0(t) \exp(\gamma \text{lowTV} + \alpha m_i(t)), \tag{2}$$

where $h_0(t)$ denotes the baseline hazard at time t , γ is the change in the log hazard of death over 28 days of resulting from low tidal volume, controlling for the trajectory of IL-6 concentration. $\exp(\gamma)$ gives the hazard ratio. α is the change in the log hazard of death over 28 days for a one log unit change in the estimated longitudinal trajectory $m_i(t)$. $\exp(\alpha)$ is therefore the hazard ratio for a one unit increase in IL-6.

3.3 Visual representation of the mediation model



3.4 Direct and Indirect Effects

We fit the above joint models in the R package JMBayes2 [2] and use the obtained estimates to compute direct and indirect effects in two ways: (1) as hazard ratios and (2) as cumulative risk differences. Hazard ratios are convenient estimands because they provide point estimates for the multiplicative total, direct, and indirect effects of intervention on survival outcomes. However, hazard ratios are well known to be non-collapsible, which means that here, the direct and indirect hazard ratios do not sum to the total hazard ratios. Therefore, inferences about *how much* of a total intervention effect is mediated based on hazard ratios are non-viable. Cumulative risk differences offer a more intuitive understanding of total, direct, and indirect effects over time. We use these estimates to visualize relative magnitudes.

3.4.1 Direct and indirect hazard ratios

The indirect hazard ratio is $\exp(\beta_1 \times \alpha)$, where β_1 represents the effect of low tidal volume on IL-6 over time, and α is the estimate of the association between the value of IL-6 concentration and the hazard. The direct hazard ratio is $\exp(\gamma)$, which represents the effect of the intervention on survival, adjusting for the longitudinal mediator IL-6. The presentation of direct and indirect hazard ratios obtained using joint models aligns with the approach by Oudenhoven et al. [2].

We compute Bayesian credible intervals to quantify uncertainty around the estimates of indirect and direct hazard ratios. For this, we use the MCMC samples of the estimates for β_1 , α , and γ . The credible interval for the indirect effect is obtained by multiplying the MCMC samples β_{1m} and α_m at each iteration m .

$$\text{IND}_m = \beta_{1m} \times \alpha_m$$

We then take the 2.5th and 97.5th percentiles of IND_m to obtain 95% credible intervals.

$$\text{CI}_{2.5\%} = \text{quantile}(\text{IND}_m, 0.025)$$

$$\text{CI}_{97.5\%} = \text{quantile}(\text{IND}_m, 0.975)$$

3.4.2 Direct, indirect, and total effects as cumulative risk differences

To obtain direct and indirect effects on a cumulative risk difference scale, we apply the concept of "treatment separation" [3]. Treatment separation is the assumption that there is an aspect of treatment that affects the survival outcome directly and a separate aspect of treatment that affects the survival outcome only via the mediator. Under this assumption, we can simulate scenarios where we "intervene" on only one

aspect of treatment and obtain risk estimates. Below we illustrate the steps used to compute direct, indirect, and total effects on a risk difference scale using joint modeling. We continue using the intervention with low tidal volume ventilation and IL-6 as a mediator as an example. Note that this is a g-computation approach, where we artificially intervene on an exposure to estimate counterfactual outcomes under different settings.

1. Create a second treatment variable $lowVT^D$ that is equivalent to the original treatment variable $lowVT$. We rename the original treatment variable into $lowVT^M$. The superscript D denotes "direct" and M denotes "indirect".
2. Fit the joint model as defined previously, now using the treatment variable $lowVT^D$ in the survival submodel and $lowVT^M$ in the mixed effects submodel. The output of this joint model is identical to the previously defined joint model where we used $lowVT$ in both submodels.
3. Clone the data and set both treatment variables to 1 (i.e., intervention with low tidal volume). Corresponds to $do(lowVT^D = 1, lowVT^M = 1)$.
4. Clone the data and set both treatment variables to 0. Corresponds to $do(lowVT^D = 0, lowVT^M = 0)$.
5. Clone the data and set the aspect of treatment that affects the mediator to 1 and the aspect of treatment that affects the survival outcome to 0. Corresponds to $do(lowVT^D = 0, lowVT^M = 1)$.
6. Clone the data and set the aspect of treatment that affects the mediator to 0 and the aspect of treatment that affects the survival outcome to 1. corresponds to $do(lowVT^D = 1, lowVT^M = 0)$.
7. Using the fitted joint model, estimate the mediator (here biomarker IL-6) values over time for all four data sets.
8. In the four data sets, set the mediator values to the respective mediator values obtained in the previous step.
9. Using the fitted joint model, estimate the survival outcome over time for all four data sets.

Following these steps, we obtain estimates of cumulative risks over time under four different scenarios for each patient. The indirect, direct, and total effects can be computed as the following contrasts:

- **Total cumulative risk differences:** contrast between estimated outcomes (i.e., cumulative risks) under intervention ($do(lowVT^D = 1, lowVT^M = 1)$) versus control ($do(lowVT^D = 0, lowVT^M = 0)$).
- **Direct cumulative risk differences:** contrast between estimated outcomes under intervention on the aspect of tidal volume that affects survival directly ($do(lowVT^D = 1, lowVT^M = 0)$) versus no intervention ($do(lowVT^D = 0, lowVT^M = 0)$).
- **Indirect cumulative risk differences:** contrast between estimated outcomes under intervention on the aspect of tidal volume that affects the mediator IL-6 ($do(lowVT^D = 0, lowVT^M = 1)$) versus no intervention ($do(lowVT^D = 0, lowVT^M = 0)$).

4 Estimation details

We use a Bayesian approach to estimate the parameters of the joint model defined above. This approach employs Markov-Chain Monte-Carlo (MCMC) methods to estimate the posterior distribution of parameters. Given the observed data (IL-6 measures, survival times, and covariates), the formulation of the joint posterior distribution is possible under the assumption that the longitudinal observations of IL-6 are independent from the event times given the random effects.

Given the observed event times T_i , event indicators δ_i , and IL-6 values y_i , the joint posterior distribution of our model can be formulated as

$$p(b_i, \theta_S, \theta_Y \mid T_i, \delta_i, y_i) \propto \underbrace{\prod_{i=1}^N p(T_i, \delta_i \mid b_i, \theta_S)}_{\text{Survival likelihood}} \times \underbrace{\prod_{i=1}^N p(y_i \mid b_i, \theta_Y)}_{\text{Longitudinal likelihood}} \times \underbrace{p(b_i \mid \theta_Y)p(\theta_Y)p(\theta_S)}_{\text{Priors}}, \quad (3)$$

where $p(\cdot)$ is a probability density function, $\theta_Y = (\beta_0, \beta_1, \beta_2, \sigma^2)$ is a parameter vector of the longitudinal submodel, $\theta_S = (\gamma, \alpha, h_0)$ is a parameter vector of the survival submodel, and $b_i = (b_{i1}, b_{i2})$ is the vector of patient-specific random effects.

For the longitudinal submodel, the gaussian likelihood is given by

$$\prod_{i=1}^N p(y_i \mid b_i, \theta_Y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{1}{2\sigma^2} \sum_{j=1}^{n_i} (y_{ij} - x_{ij}^\top \beta - z_{ij}^\top b_i)^2 \right), \quad (4)$$

where the subscript j denotes the observation time for patient i , x_{ij} is the covariate vector, and z_{ij} a design vector of the random effects.

The survival likelihood is given by

$$\prod_{i=1}^N p(T_i, \delta_i \mid b_i, \theta_S) = \left[h_0(T_i) \exp(w_i^\top \gamma + \alpha m_i(T_i)) \right]^{\delta_i} \times \exp(-H_i(T_i \mid b_i)) \quad (5)$$

$$H_i(T_i \mid b_i) = \int_0^{T_i} h_0(s) \exp(w_i^\top \gamma + \alpha m_i(s)) ds,$$

where $H_i(T_i \mid b_i)$ is the cumulative hazard up to event time T_i .

We use default settings and priors from the JMBayes2 R package. The baseline hazard is approximated using 9 equidistantly placed knots for the follow up period and a weakly informative prior for the smoothing parameter, which allows for a flexible baseline hazard. By default, the penalty is on the second derivative, which means it penalized acceleration in the hazard curve. For further details please refer to JMBayes2 documentation.

The formulation of the joint posterior shows that our estimates of the indirect and direct effects do not have a marginal interpretation because they are conditional on the random effects.

References

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